

Detector-coupled Monte Carlo estimate of background and unresolved-line sensitivity for a balloon-borne 511 keV Laue-lens TES telescope

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Abstract Background determines the sensitivity of a balloon-borne 511 keV Laue-lens telescope with a transition-edge-sensor (TES) microcalorimeter. We developed an end-to-end Geant4/MEGAlib model transporting focused signal, all eight prompt families, production-position-sampled activation decays, and an explicit atmospheric 511 keV line through one detector–cryostat geometry. All streams share a 510.58–511.42 keV window, active anticoincidence, and topology/field-of-view selection. In the mass-complete reference, the selected prompt-plus-delayed diagnostic background was $(4.643 \pm 0.543) \times 10^{-2} \text{ s}^{-1}$; this differently scoped reference budget is used to locate sources, not as a like-for-like measure of shield gain. Positrons and neutrons supplied 96.8% of the prompt component, and cold-copper ^{64}Cu and ^{61}Cu accounted for 24 and two of 26 delayed records. This source map led to a directionally graded BGO shield with 40, 30, and 10 mm side, bottom, and top coverage and an open optical aperture. After a 420 eV FWHM pixel-level response convolution, the active veto reduced the prompt rate from 2.849×10^{-2} to $4.072 \times 10^{-3} \text{ s}^{-1}$. At the day-15 reference state, complete selection gave background and signal rates of 6.965×10^{-3} and $9.862 \times 10^{-4} \text{ s}^{-1}$ at a top-of-atmosphere flux of $10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$. A fixed-45°-elevation, 20 d trajectory fold yielded a central counting significance of 15.20 and a 3σ flux threshold of $1.974 \times 10^{-5} \text{ ph cm}^{-2} \text{ s}^{-1}$. The conditional component-wise transport-counting endpoint gave 5.67 and $5.293 \times 10^{-5} \text{ ph cm}^{-2} \text{ s}^{-1}$; it is a finite-transport diagnostic, not full 95% coverage.

Keywords Gamma-ray instrumentation, 511 keV annihilation line, transition-edge sensors, Laue lens, balloon-borne telescope, activation background

1 Introduction

Galactic 511 keV annihilation radiation is a key observational probe of the production, propagation, and annihilation environments of Galactic positrons. Its spectrum contains a two-photon line near 511 keV and an ortho-positronium continuum at lower energies. Positrons annihilate directly with electrons or form para-positronium, producing two photons near 511 keV, whereas ortho-positronium decays into a three-photon continuum. The line intensity, profile, and positronium continuum therefore constrain the medium in which positrons slow down and annihilate [1, 2]. Under an approximate steady-state assumption, the total Galactic 511 keV flux measured by INTEGRAL/SPI corresponds to an annihilation rate of a few $\times 10^{43} \text{ e}^+ \text{ s}^{-1}$ [1, 3]. Proposed positron sources include β^+ -unstable nuclei synthesized in stellar explosions, compact objects capable of producing electron–positron pairs, and more speculative particle-physics scenarios in which annihilating or decaying light dark matter injects low-energy positrons. Existing observations do not uniquely determine the dominant contribution [1, 4, 5].

Wide-field observations have established the diffuse morphology of the line. INTEGRAL/SPI finds a bright central bulge and a weaker disk, and a recent 20-year analysis recovers central, broad-bulge, and disk components while the evidence for additional correlated structures remains marginal [4, 6, 7]. The 2016 COSI balloon flight independently detected the bulge and found evidence that the emission is more extended than a single point source [3, 8]. Spectroscopy also constrains the annihilation medium and kinematics: the SPI spectrum can be decomposed into a narrow component with about 1.3 keV FWHM and a broad component with about 5.4 keV FWHM, and spatial variations in centroid energy and width have been sought across the inner Galaxy [2, 9]. The

annihilation map, however, records where positrons slow down and annihilate, not necessarily where they were created. Propagation through a magnetized, multiphase interstellar medium means that diffuse morphology and spectroscopy alone cannot recover the dominant source class or the complete pre-annihilation transport history [1].

Wide-field Compton and coded-mask instruments are well suited to the total bulge and disk emission. A complementary pointed instrument is needed to test whether the central region also contains an unresolved central enhancement, compact sources, or measurable line-kinematic structure. A detection of a point-like or narrowly concentrated feature would show that part of the annihilation is spatially localized; a non-detection would constrain a compact component without excluding source populations whose positrons escape and annihilate diffusely. An INTEGRAL/IBIS compact-source search found no Galactic 511 keV point sources and set a 2σ upper limit of about $1.6 \times 10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$ for a ~ 10 Ms Galactic-centre exposure [10]. This limit sets both the comparison scale of earlier compact-source searches and a sensitivity target for a new narrow-field pointing instrument.

Laue optics offer one route to such pointed observations by using transmission-mode Bragg diffraction to focus photons from a known direction onto a small focal plane [11, 12]. Concentrating the flux collected over a large aperture onto a smaller sensitive detector area can improve point-source sensitivity when detector-related background dominates [13]. The focal plane must then combine a narrow energy response with sufficient stopping efficiency at 511 keV. Transition-edge-sensor (TES) microcalorimeters use the steep resistive transition of a superconductor to measure small temperature changes from deposited energy and can provide sub-keV resolution at 511 keV [14, 15]. An array covers the focal spot, while thick absorbers and a multilayer layout raise the full-energy efficiency. Laue focusing compresses the accepted phase space, and the TES response permits a narrow line window and detailed line-profile measurements. The resulting instrument trades field of view for a concentrated beam and high-resolution spectroscopy, complementing rather than replacing wide-field bulge and disk measurements.

However, the intrinsic high energy resolution of the TES detector and the focusing performance of the optics are not by themselves sufficient to establish the point-source resolving capability of the complete detection system [14, 16]. Both the prompt background spectrum generated by cosmic-ray energy deposits in the detector's sensitive volume and the secondary delayed background spectrum produced by cosmic-ray activation of the detector and surrounding materials affect the system's ability to resolve a scientific source [17, 18].

This work aims to quantify the high-altitude detection performance of a balloon-borne Laue-lens TES-array telescope system for a 511 keV point source at the Galactic center, while characterizing the origin and distribution of the background to guide further design optimization.

1.1 Science and instrument requirements

The simulated instrument is designed to detect either 511 keV point-source emission or an unresolved central-excess component in the Galactic-center region, neither of which has been detected by previous instruments. INTEGRAL/SPI measurements show that the 511 keV sky is dominated by diffuse bulge and disk emission [4, 7]. De Cesare's dedicated search for compact sources with INTEGRAL/IBIS found no Galactic 511 keV point sources and reported a 2σ upper limit for a 10 Ms exposure toward the Galactic center [10]. To compare this limit with the 3σ sensitivity reported here, we linearly rescale the published limit under the Gaussian approximation, obtaining an IBIS-based 3σ -equivalent top-of-atmosphere upper limit of $2.4 \times 10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$. At the altitude on flight day 15, the remaining vertical atmospheric column depth is 3.461 g cm^{-2} . For the instrument axis, designed for a fixed elevation of 45° , the plane-parallel approximation gives a slant column depth of 4.895 g cm^{-2} ; using an effective dry-air mass attenuation coefficient of $0.08736 \text{ cm}^2 \text{ g}^{-1}$, consistent with NIST XCOM data near 511 keV, gives a transmission of $T_{15,45} = 0.6520$. Applying this transmission defines the single observational reference flux at the payload plane, $F_{0,\text{ref}} = (2.4 \times 10^{-4}) \times 0.6520 = 1.565 \times 10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$. This reference flux provides a benchmark for the minimum resolvable flux achievable by the simulated telescope system within an acceptable flight duration, as reported below. Compared with the wide-field spectroscopic and coded-mask searches conducted with INTEGRAL, the payload modeled here introduces Laue focusing to improve the point-source response and uses a high-energy-resolution, multilayer thick-absorber TES microcalorimeter array as the focal-plane detector to improve its capability for 511 keV line measurements [14, 19].

This work first considers a balloon payload as an early test platform, with the expectation that the approach can later be extended to a satellite observing platform.

This objective imposes four requirements on the instrument and the simulation. First, the telescope must focus the 511 keV photons collected by a large-aperture lens onto a compact focal plane, thereby increasing the ratio of the collection area of the focusing optics to the sensitive area of the detector; this is a narrow-field pointed configuration, not a means of increasing sky coverage. Second, the focal-plane detector must provide a sub-keV energy response sufficient to resolve fine line structure. Third, the detector and shielding model must suppress atmospheric, activation, and side-entry backgrounds. Fourth, the simulation chain must treat prompt background, delayed activation, focused signal, vetoes, event topology, and mission-time variation consistently, so that signal and background are evaluated using the same detector-selection criteria.

The remainder of this paper is organized as follows. Section 2 presents the detector and cryostat mass model; Section 3 describes the simulation framework, including the simulation workflow, optics coupling, background sources, event selection, and mission-time trajectory folding (Section 3.5); Section 4 presents the post-selection rates and sensitivity results together with the normalization, convergence, and boundary checks; and Section 5 contains the discussion and conclusions.

2 Detector and cryostat mass model

This study uses a generic mass-model template developed during an earlier stage of the project. It contains a balloon-borne multilayer TES array mounted at the cold end of a compact dilution refrigerator (DR), together with cold plates and thermal shields at the successive temperature stages, magnetic and passive shields, side-entry windows, a collimator, copper thermal links, active scintillators, and cryogenic service structures [14,20]. Figure 1 shows a schematic of this mass-model template; the quantitative calculations distinguish the initial CsI reference geometry from the final BGO geometry described below.

The TES detector concept uses an absorber-coupled AlMn-film structure developed in our laboratory. The transition temperature of AlMn can be tuned through its composition and annealing process; the material is suitable for array fabrication and has relatively low magnetic-field sensitivity [21,22]. In the present conceptual design, the TES thermometer is an approximately 200 nm-thick AlMn film with a nominal Mn atomic concentration of 2000 ppm, whereas the γ -ray absorber is a millimetre-scale metal body. The absorber consists of $1.5 \times 1.5 \times 3.0 \text{ mm}^3$ Ta pixels; the high atomic number of Ta increases the interaction efficiency for 511 keV photons. At 511 keV, a single 3.0 mm Ta layer has an estimated interaction efficiency of about 49%, and the corresponding six-layer efficiency is close to unity. In the current geometry, adjacent Ta array layers have a center-to-center spacing of 12 mm; this spacing was chosen with a subsequent veto based on Compton-kinematic track reconstruction in mind. Ta was also selected for its low volumetric heat capacity as a superconductor at cryogenic temperatures (an order-of-magnitude estimate of $10^{-2} \text{ J K}^{-1} \text{ m}^{-3}$ is used here) and its relatively high photoelectric-to-Compton interaction ratio. From the NIST XCOM cross sections near 511 keV, this ratio is approximately 0.731 [23]. In the detector-response model, we use our laboratory experience with a previously developed X-ray TES [21] to adopt $\Delta E_{\text{FWHM}} \simeq 420 \text{ eV}$ at 511 keV as the estimated energy resolution and detector-response post-processing input.

Because the TES film and Ta absorber differ greatly in dimensions and mass, and because the TES film itself contributes negligibly to interactions at 511 keV, the transport geometry uses the Ta absorber as the principal sensitive volume of each TES pixel. The explicitly modelled sensitive volume comprises six layers, each containing 376 Ta absorber pixels, for a total of 2256 pixels. Each pixel is 1.5 mm long and wide and 3 mm thick. A Si substrate proxy is placed below each Ta pixel; because detailed thermal conduction is not modelled, Si represents the $\text{SiO}_2\text{--SiN}_x\text{--Si}$ substrate. The TES film, wiring, and readout details are not constructed layer by layer as explicit interaction volumes, but enter the model through the assumed energy response and service-mass proxies. Defined independently of the response FWHM, the retained W2 analysis selection uses the 511 keV line window $W_{511} = 511 \text{ keV} \pm 420 \text{ eV}$, or 510.58–511.42 keV.

The TES array is mounted on copper supports and thermal links. A copper thermal finger passes through the rear of the sample box and connects to the 50 mK cold plate. The near-field sample box has a 2 mm Nb inner layer and a 2 mm MuMetal outer layer; MuMetal is approximated as Ni:Fe = 4:1 in the material table. Along the optical path are successive 300 μm Al thermal-shield windows, a 25 μm Be vacuum window, and an 8 mm W multihole collimator aligned with the TES pixels. These side-entry apertures are generated by Boolean cuts

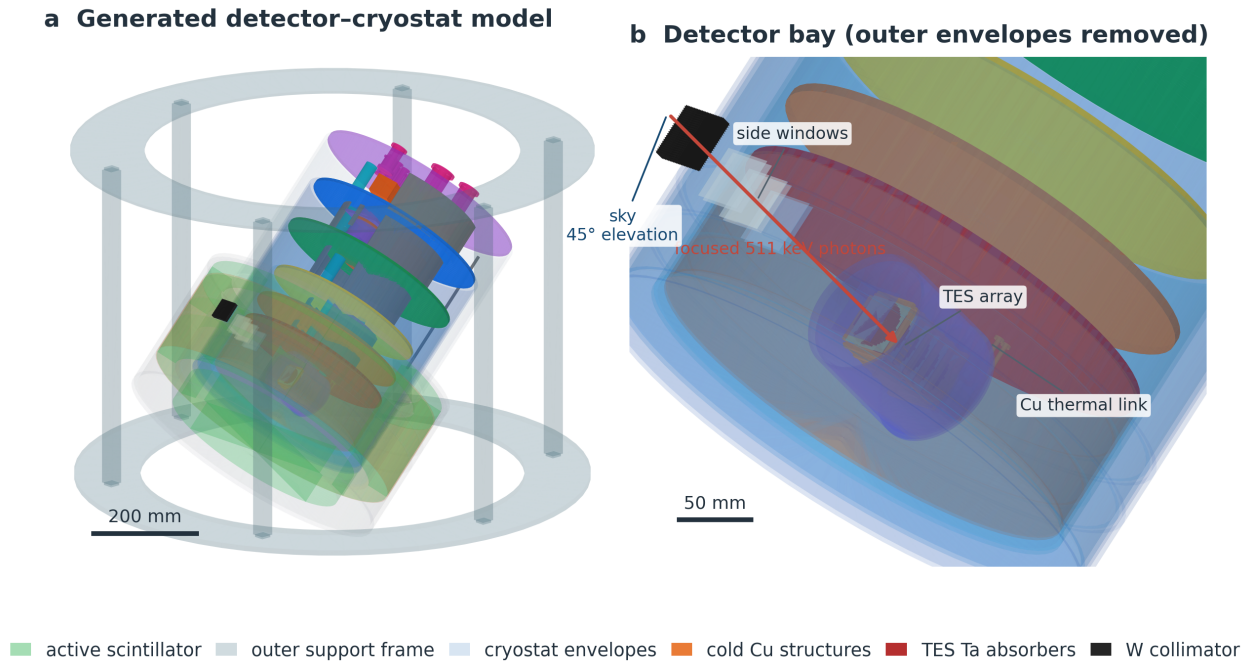


Figure 1: Schematic of the generic detector-cryostat mass-model template developed during an earlier stage of this project. (a) Overview of the template, including the outer support frame, active-scintillator shield, cryostat stages, and service structures. (b) Detector-bay view with the outer envelopes hidden to show the TES Ta array, copper thermal link, side-window stack, and W collimator. The aperture axis is oriented at 45° elevation; the red arrow indicates the nominal path of focused 511 keV photons from the sky-facing collimator to the TES array. Selected outer volumes are rendered translucent for clarity.

through the surrounding shells. The initial reference geometry used segmented CsI on the sidewalls and end surfaces; the final optimized geometry replaces this outer active shield with BGO. The entire instrument frame is rotated by 45° , which tilts the local side window upward by 45° in the zenith reference coordinates.

For the final scintillator active-veto calculation, energy deposits in the active-shield volumes are summed event by event in post-processing. The common detector-level anticoincidence selection then applies an offline deposited-energy veto threshold of 50 keV for the scintillator shield; events with a summed shield deposit at or above this threshold are vetoed.

3 Simulation framework

Detector-coupled Monte Carlo transport through detailed mass models is a mature method for predicting the response and background of gamma-ray spectrometers [24]. This section describes the end-to-end chain from source generation to flight-performance assessment. The focusing optics and detector system are represented by independent mass models. In the optics branch, the target point source is transported through the Laue-lens model, and the phase space of photons reaching the focal plane is recorded. On the detector side, four independently constructed inputs are transported through the detector-cryostat mass model of Section 2: the focal-plane EventList constructed from the focused-photon phase space, the broadband prompt atmospheric field, delayed decays sampled at their recorded production positions, and a monoenergetic atmospheric 511 keV prompt-background source constrained by published measurements. The four streams receive the same detector response and 511 keV energy-window selection, followed by the active veto and Compton-topology test. An explicit common-time-axis realization of the focused, broadband-prompt, and delayed catalogues validates the Poisson coincidence grouping; the final mission fold additionally includes the atmospheric-line rate and detector occupancy in each trajectory bin. The selected streams provide the signal rate and component-resolved background rates used for background-source decomposition and mission time-trajectory folding. These two analyses jointly inform the final shielding-geometry optimization and design comparison, while the common response and selection contract keeps all subsequent comparisons on a consistent basis.

Figure 2 organizes the chain into three physical branches: focused signal, prompt external atmospheric back-

ground, and delayed activation background. Within the prompt branch, the EXPACS/PARMA broadband field and the literature-constrained monoenergetic atmospheric 511 keV line are transported independently through the complete detector–cryostat mass model. The resulting event catalogues are subjected to the same detector response, active veto, and Compton-topology test before the selected rates enter background-source decomposition and mission time–trajectory folding. The resulting source and mission-performance information is then used for the final shielding-geometry optimization and design comparison. The following subsections describe the inputs, operations, and outputs of these stages.

Day-scale flights sample a continuum of altitude and geomagnetic states, so a single Monte Carlo snapshot cannot be linearly stretched in time. The mission-time trajectory fold—its motivation, analytic definitions, multi-point transport tests, and the resulting performance curves—is developed in Section 3.5.

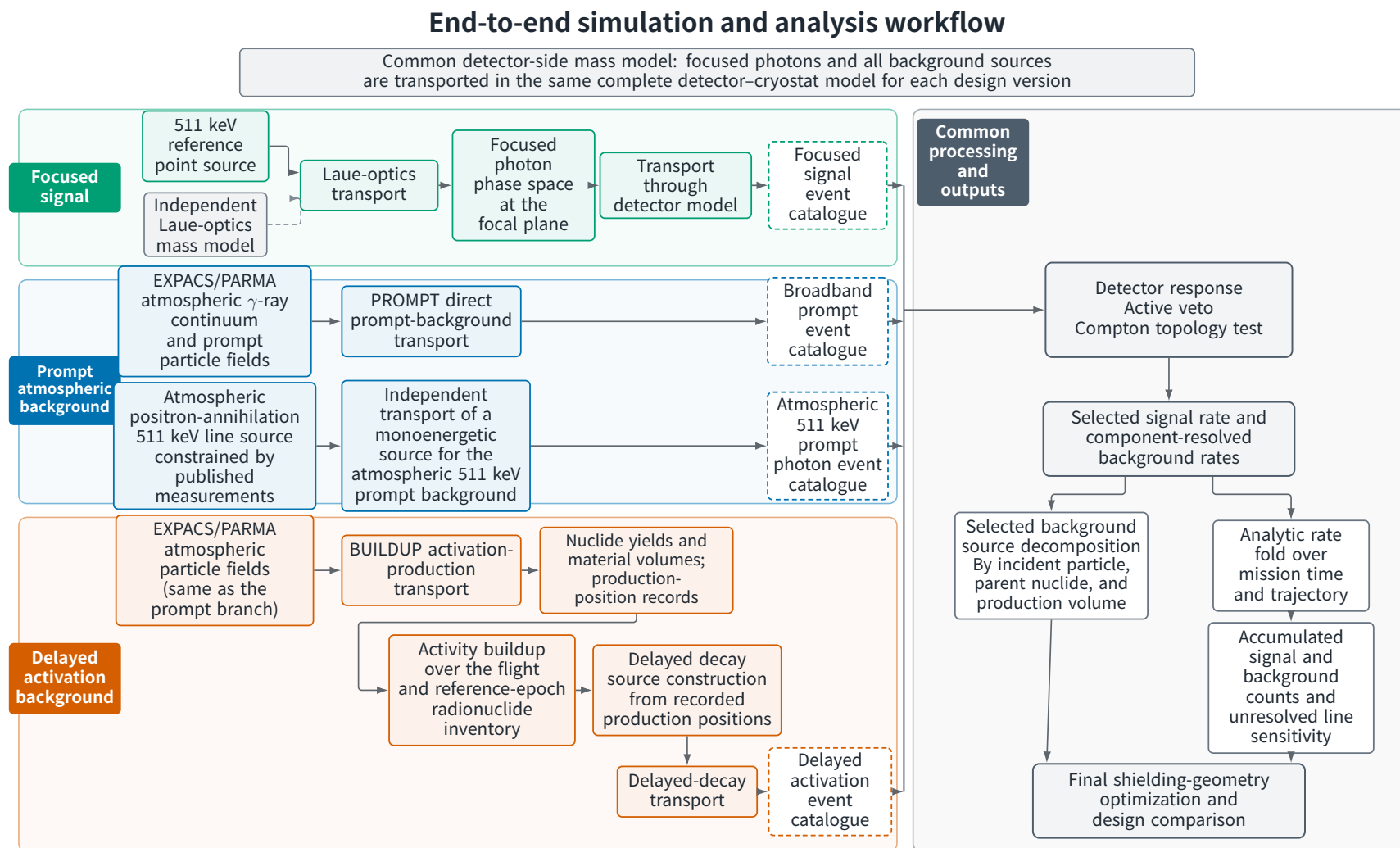


Figure 2: End-to-end simulation and analysis workflow. The focused signal, two prompt atmospheric streams, and delayed activation background are transported separately and then subjected to a common detector response, active veto, and Compton-topology test. The selected rates enter background-source decomposition and mission time–trajectory folding, which jointly inform the final shielding-geometry optimization and design comparison.

3.1 Laue optics and detector-coupled signal transport

This simulation adopts a laue-lens system based on mosaic Ge(111) crystals [11, 12]. Through optical design, it realizes a 10 m focal length, a 511 keV single-ring reference response, and an effective area of 20.1 cm² at 511 keV; the effective area is calculated as

$$A_{\text{eff}}(E) = A_{\text{inc}} \frac{N_{\text{fp}}(E)}{N_{\text{inc}}(E)} \quad (1)$$

where A_{inc} is the Monte Carlo incident sampling area, $N_{\text{inc}}(E)$ is the number of incident photons, and $N_{\text{fp}}(E)$ is the number of photons satisfying the focusing/focal-plane interception condition. The mosaic system is adopted to make the Monte Carlo calculation convenient while satisfying the expected focusing performance; other candidate schemes are also possible for actual engineering adaptation.

In this simulation, the laue optical branch gives the focusing-performance simulation and transports the focused signal through the Ge crystal mass.

For this purpose, the diffraction probability of photons by the crystal is no longer superposed onto photons as an efficiency parameter, but is defined as a discrete process at the Geant4 application layer [25, 26]. The process takes the diffraction probability $R(|\Delta\theta|)$ according to the angular deviation of the current photon relative to the crystal plane, and converts it into a finite equivalent diffraction mean free path λ_{diff} , so that the accumulated diffraction probability when the photon crosses a crystal path ℓ reproduces this value: $1 - \exp(-\ell/\lambda_{\text{diff}}) = R(|\Delta\theta|)$, namely $\lambda_{\text{diff}} = -\ell / \ln[1 - R(|\Delta\theta|)]$; to avoid double counting with standard photoelectric absorption, R is first normalized by the unabsorbed fraction ($R/(1 - A)$) and then converted into a mean free path. After this mean free path is returned to Geant4, the diffraction process and the standard electromagnetic processes (photoelectric absorption and Compton scattering) each sample interaction lengths in the same competition framework, and the shortest one decides which process occurs first, rather than forcibly intercepting photons at the crystal boundary. Photons not selected to undergo diffraction still continue transport according to the standard electromagnetic physics, and may be absorbed, scattered, or directly transmitted.

The diffraction probability is preferentially given by interpolation in an external XOP/CRYSTAL rocking-curve map as a function of angular deviation: this curve is calculated offline once in the preprocessing stage by the CRYSTAL/diff_pat module of the XOP toolkit (called through xoppylib, with DABAX crystallographic data as input) [27], and gives the Ge(111) reflectivity/transmissivity/absorption curves as a function of angular deviation; during transport, the angular deviation is calculated in real time for every step inside the crystal, and the current diffraction probability is obtained by interpolation on that curve, namely “offline library tabulation, probability used by angle during transport”. The mosaic orientation is implemented by applying a Gaussian angular perturbation to the ideal Bragg-plane normal, with the perturbation width given by the 30 arcsec mosaic FWHM; after diffraction occurs, the outgoing direction is mirror-reflected according to the perturbed plane normal, and the photon energy is unchanged. This split in which the “process class only handles the transport interface, and the independent model class handles probability and direction decisions” follows the idea of existing crystal Bragg-reflection extensions in Geant4 [28, 29], and extends it to the energy band needed here.

Here the reference signal is a monochromatic 511 keV point source whose rounded top-of-atmosphere flux scale ($F_0 = 10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$, Section 1.1) is motivated by the dedicated INTEGRAL/IBIS compact-source upper limit [10], rather than by any single observed object. It is constructed as a MEGAlib far-field source [30], with the detailed construction given in Section 3.2.4. The signal is transported through the optics branch and injected into the detector mass model as a parallel-beam source; prompt cosmic-ray background and activation decays are transported directly in the detector/cryostat mass model. The primary background budget is therefore defined for the detector/cryostat branch, while the optics branch supplies the transported focal-plane signal phase space.

3.2 Source construction

After the focusing optics have been defined in Section 3.1, four transport components are constructed independently: the prompt atmospheric particle field incident on the payload, the delayed radioactive population accumulated in the instrument, a monochromatic atmospheric 511 keV line, and the reference signal point source. Activation production and delayed-decay transport remain separate stages. This construction preserves the normalization and provenance of each component before the common detector response and event selection are

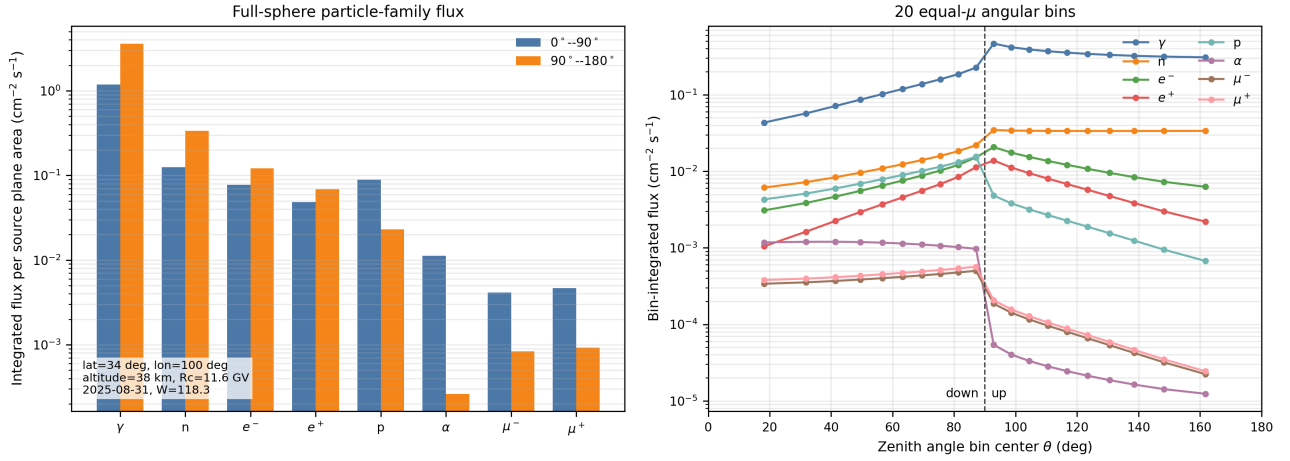


Figure 3: EXPACS/PARMA full-sphere atmospheric flux field used to generate the prompt and activation-production transport inputs. Left: energy- and solid-angle-integrated fluxes for the eight transported particle families, separated into down-going and up-going hemispheres. Right: bin-integrated flux in the 20 equal- μ differential zenith bins used by the MEGALib far-field source. The plot is generated from the source-definition inputs for latitude 34° , longitude 100° , altitude 38 km, $R_c = 11.6$ GV, and $W = 118.3$.

applied.

3.2.1 Prompt atmospheric source

The prompt atmospheric field is derived from EXPACS/PARMA. PARMA3.0 is an analytical model parameterized from PHITS atmospheric-shower calculations and benchmarked against terrestrial cosmic-ray measurements, and EXPACS is its public tabulation and software interface [31, 32]. PHITS supplies the underlying particle-transport physics: primary cosmic rays interact with air nuclei, the secondary hadrons, leptons, photons, and neutrons are transported through the atmospheric column, and the resulting fluences are parameterized by altitude, geomagnetic cutoff, solar modulation, energy, and direction. In this work EXPACS/PARMA provides the differential incident flux as a function of particle species, kinetic energy, zenith angle, geographic position, altitude, cutoff rigidity, and solar activity, and is used only to construct the external radiation fields; detector response, vetoing, topology selection, and mission-time counting are applied downstream. The reference source definitions use latitude 34° , longitude 100° , altitude 38 km, vertical cutoff rigidity $R_c = 11.6$ GV, and the 2025-08-31 solar condition corresponding to $W = 118.3$.

The prompt source includes photons, neutrons, electrons, positrons, protons, alpha particles, and negative and positive muons. For each particle family the EXPACS/PARMA differential spectrum is converted into a MEGALib far-field area source covering the full zenith range $\theta = 0^\circ\text{--}180^\circ$ and full azimuth range $\phi = 0^\circ\text{--}360^\circ$. The zenith dependence is represented by 20 differential equal- μ angular bins, where $\mu = \cos \theta$ and each bin subtends $\Delta\Omega = 0.6283185307$ sr: the first ten bins cover down-going particles, the last ten up-going or albedo-like particles. Both hemispheres are retained in the prompt transport and in the activation-production transport, so upward albedo particles are not discarded at source construction. Figure 3 shows the resulting full-sphere flux field.

The integrated source-plane fluxes are approximately 4.80, 0.462, 0.199, 0.117, 0.112, 0.0115, 0.0050, and 0.0056 $\text{cm}^{-2} \text{s}^{-1}$ for γ , n , e^- , e^+ , p , α , μ^- , and μ^+ , respectively; the non-photon families together carry about 0.91 $\text{cm}^{-2} \text{s}^{-1}$, or 16% of the total. If Monte Carlo counts were allocated strictly in proportion to flux, the detector and activation statistics of these minority families would be poor, so the non-photon families are generated with replicated samples and de-weighted in the rate normalization. All particles are then transported through the detector/cryostat mass model with Geant4/MEGALib [25, 26, 30]; the prompt run and the activation-production run each generated 25,210,216 incident primaries. Each particle species is normalized with its own

transported exposure,

$$r_{\text{event},j} = \left(\sum_i \mathcal{T}_{ij} \right)^{-1}, \quad (2)$$

where $\mathcal{T}_{ij}$ is the transported exposure of subsample i for species j . This per-species normalization is required because the photon and non-photon streams have different sampling and exposure structures; the normalization check enforces $r_{\text{event},j} \sum_i \mathcal{T}_{ij} = 1$ for every species, so over-sampling changes the Monte Carlo variance but not the expected physical rate.

3.2.2 Atmospheric 511 keV line source

The 20 atmospheric photon tables used for the EXPACS/PARMA prompt source represent a smooth continuum. In every angular bin the tabulated energy grid has adjacent entries at 449.65 and 566.08 keV, so the value at 511 keV is obtained by continuum interpolation and not by an independent line entry. We retained these tables unchanged and transported a separate atmospheric positron-annihilation component modeled as monoenergetic at 511 keV; detector broadening is applied in the subsequent response stage. This numerical separation does not establish that the smooth continuum is free of unresolved annihilation-related emission. Possible overlap is therefore recognized as an unquantified source-model uncertainty rather than assumed to be zero.

The line source is semi-empirical. Its reference normalization, rigidity factors, and upward limb-darkening form are informed by atmospheric gamma-ray measurements [33, 34]; the depth exponent, hemispheric-ratio parameters, and downward angular kernel are modeling assumptions. The full-sphere line-integrated flux is

$$\Phi_{4\pi}(X, R_c) = \Phi_{\text{ref}} f_R(R_c) \left(\frac{X}{X_{\text{ref}}} \right)^\eta f_\odot, \quad (3)$$

where the baseline uses $\Phi_{\text{ref}} = 0.10 \text{ ph cm}^{-2} \text{ s}^{-1}$, $X_{\text{ref}} = 3.6 \text{ g cm}^{-2}$, $\eta = 0.8$, and $f_\odot = 1$. The adopted rigidity factor is the step function

$$f_R(R_c) = \begin{cases} 1.000, & R_c < 7 \text{ GV}, \\ 0.805, & 7 \leq R_c < 9 \text{ GV}, \\ 0.624, & 9 \leq R_c < 11 \text{ GV}, \\ 0.532, & 11 \leq R_c < 13 \text{ GV}, \\ 0.479, & R_c \geq 13 \text{ GV}. \end{cases} \quad (4)$$

The downward-to-upward flux ratio and hemispheric fluxes are

$$r_\downarrow(X) = \min \left[1, r_0 \left(\frac{X}{X_{\text{ref}}} \right)^\beta \right], \quad \Phi_\uparrow = \frac{\Phi_{4\pi}}{1 + r_\downarrow}, \quad \Phi_\downarrow = \frac{r_\downarrow \Phi_{4\pi}}{1 + r_\downarrow}, \quad (5)$$

with $r_0 = 0.20$ and $\beta = 0.5$.

Here θ is the source zenith angle: $\theta = 0$ denotes photons arriving from overhead and $\theta = \pi$ denotes upward/albedo photons. For the upward hemisphere, $\mu_\uparrow = -\cos \theta$ and the normalized limb-darkening law is

$$I_\uparrow(\mu_\uparrow) = \frac{\Phi_\uparrow}{2\pi(1 + a/2)} (1 + a\mu_\uparrow), \quad a = 1.7. \quad (6)$$

For the downward hemisphere, $\mu_\downarrow = \cos \theta$ and the adopted slab-production/escape kernel is

$$\begin{aligned} K_\downarrow(\mu_\downarrow, X) &= 1 - \exp \left[-\frac{X}{\Lambda_{511}\mu_\downarrow} \right], & \Lambda_{511} &= 11.5 \text{ g cm}^{-2}, \quad 0 < \mu_\downarrow \leq 1, \\ I_\downarrow(\mu_\downarrow, X) &= \frac{\Phi_\downarrow}{2\pi\mathcal{A}(X)} K_\downarrow(\mu_\downarrow, X), & \mathcal{A}(X) &= \int_0^1 K_\downarrow(\mu, X) d\mu. \end{aligned} \quad (7)$$

The continuous limit $K_{\downarrow}(0, X) = 1$ is used at the horizon. The attenuation scale is consistent with the dry-air mass attenuation coefficient near 511 keV [23].

The full sphere is discretized into the same 20 equal- μ bins used by the prompt source, with 10 bins in each hemisphere, $\Delta\mu = 0.1$, and $\Delta\Omega = 2\pi\Delta\mu = 0.628319$ sr. MEGAlib samples the far-field area sources on the projected disk associated with a 60 cm start sphere centered at (5, 0, 9) cm; the normalization uses $A_{\text{proj}} = \pi R^2 = 1.13097 \times 10^4 \text{ cm}^2$ rather than the physical surface area $4\pi R^2$. At the day-15 proxy state used by the line-transport source card, $X = 3.46147 \text{ g cm}^{-2}$ and $R_{c,\text{line}} = 11.0 \text{ GV}$ give $\Phi_{4\pi} = 0.0515559 \text{ ph cm}^{-2} \text{ s}^{-1}$, with $\Phi_{\uparrow} = 0.0431028$ and $\Phi_{\downarrow} = 0.00845307 \text{ ph cm}^{-2} \text{ s}^{-1}$. The Monte Carlo transport generated 3.0×10^6 photons, corresponding to an effective exposure of 5143.83 s and a per-event rate weight of $1.94408 \times 10^{-4} \text{ s}^{-1}$. The state-specific PARMA field used for prompt and activation scaling has $R_{c,\text{PARMA}} = 11.6 \text{ GV}$ at day 15. Both values lie in $11 \leq R_c < 13 \text{ GV}$ and therefore use the same $f_R = 0.532$; the atmospheric-line branch retains its separate proxy rigidity curve in the mission fold.

3.2.3 Delayed activation source

For the delayed simulation, a separate activation-production stream is run by enabling BUILDUP during the prompt-background transport. This activation-production stream is not counted as an immediate detector background. Instead, it records the isotope, material volume, and production position of each radioactive product. These records are converted through the production-decay accumulation equation into a radioactive population fixed to a chosen reference time during the flight, with ground-state half-lives checked against NUBASE2020 [35]. For isotope k ,

$$A_k(t_{\text{flight}}) = P_k [1 - \exp(-\lambda_k t_{\text{flight}})], \quad \lambda_k = \frac{\ln 2}{t_{1/2,k}}, \quad (8)$$

where P_k is the production rate inferred from the activation-production transport and $t_{\text{flight}} = 15 \text{ d}$ for the reference state. The fixed delayed total activity used by the mass-complete reference delayed-source construction is 141.83 Bq. This reference value is retained for the background-origin comparison below; the final directionally graded chain has its own all-eight-family inventory and normalization.

The position information does not come from the aggregate isotope store alone. The standard store records volume names and isotope counts; the production-position construction adds a custom recording hook in the MEGAlib stepping-action layer. Whenever a delayed isotope is stored during activation buildup, the hook writes a production-position record at the same moment, including the logical volume name, pre-step position (x, y, z), isotope identifier $1000Z + A$, excitation energy, time, process, and track-ancestry information. The fixed day-15 population is then matched to these records by volume, isotope, and excitation state, so the delayed source can use the simulated production coordinates rather than only volume-integrated activities. The reference-detector inventory is summarized by production volume, nuclide, incident family, material category, and production coordinate. These are source-construction quantities rather than detector counts; detector response is applied only after the delayed decays are transported through the common geometry and event selection.

Whether the production positions are worth preserving can be tested from the same table. Delayed decays are emitted locally, so downstream attenuation, active-shield veto probability, and event topology all depend on where the isotope was produced, and different incident families—with their different energy-loss, stopping, and capture histories—need not deposit their activation in the same materials. To quantify this, the weighted production rows are grouped by incident particle family and material category, and the activity-normalized category distributions of two families a and b are compared with the total-variation distance

$$D_{\text{TV}}(a, b) = \frac{1}{2} \sum_c |f_{a,c} - f_{b,c}|, \quad (9)$$

where $f_{a,c}$ is the fraction of delayed activity from family a produced in material category c . Production volumes are grouped as active CsI, outer mechanical structures, cold plates, passive tungsten/collimator material, window material, TES volumes, and other internal structures. A volume-integrated or axisymmetric radial source could preserve the total activity, but it would erase these incident-family, material, nuclide, and coordinate correlations; the production-position construction keeps them in the source before detector transport.

The delayed-decay source is therefore emitted from sampled production positions rather than from an axisymmetric distribution. Let row j in the weighted production table have activity weight w_j , with total activity $A = \sum_j w_j$. During construction, M production positions are drawn with replacement using probability $p_j = w_j/A$, and each sampled point source is assigned the same flux A/M . The expected activity assigned to row j is

$$\mathbb{E}\left[N_j \frac{A}{M}\right] = M p_j \frac{A}{M} = w_j, \quad (10)$$

so the estimator preserves the total activity and is unbiased for the weighted spatial distribution. Here M is the number of equal-activity sampled point sources; it is a sampling size, not the number of physical production rows. The reference delayed source preserves the fixed total activity at the source-definition level and is transported through the same detector/cryostat geometry as the prompt stream. The nominal calculation transports 10^6 delayed decays; the corresponding Cosima live time is only the transport record used to generate the decay sequence and is not used as the physical balloon exposure time.

Table 1 summarizes the source layers used in the current detector-coupled calculation. It is placed in the source-model section because this section defines incident particle fields and delayed radioactive populations; detector vetoes and topology selections are applied only in the subsequent selection section.

This production-position representation does not create one sampled point source for every production row. Each positive-family realization uses $M = 50,000$ sampled positions and transports 10^6 delayed triggers. We first verify activity conservation and source serialization, then propagate the same realization through the detector response and event selection. The resulting selected rate and its nuclide/volume decomposition are reported in Section 4.1.

The 141.83 Bq all-family day-15 inventory above is used to identify the activation origins of the mass-complete reference geometry, whose 26 selected delayed records remain 24 ^{64}Cu and two ^{61}Cu records. The final graded-shield calculation instead uses the NUBASE-corrected all-eight-family inventory of 36.4525153 Bq at day 15. The seven positive activation families with delayed transports are α , e^+ , γ , μ^- , μ^+ , n , and p ; the finite BUILDUP sample for e^- observed no production, so its central delayed contribution is zero and no transport exposure or Garwood bound is invented. The primary response realization retains 78 delayed records after the complete detector selection (1 α , 5 μ^- , 60 n , and 12 p ; the other positive transported families retain zero final records).

The finite M sampling is an additional source-level boundary rather than a Poisson counting interval. In the dominant neutron realization, undrawn low-probability nuclide–volume species carry 0.157388 Bq, or 0.515% of the neutron activity. This missed sampled-species fraction is reported explicitly and is not covered by the componentwise Garwood endpoint. One retained neutron event also begins in the SIM record as the daughter ^{182}W ; its production position uniquely maps it back to the radioactive source parent ^{182}Ta . Mission-time activity and response are therefore keyed to the ^{182}Ta parent, while the ^{182}W IA INIT identity is retained as transport-lineage evidence.

3.2.4 Reference point-source construction

The focused signal stream is defined by a monochromatic 511 keV reference point source. Its flux scale is set not by any single observed object but by a rounded comparison to the dedicated INTEGRAL/IBIS compact-source limit: $F_0 = 10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$ at the top of the atmosphere (Section 1.1) [10].

Because such a source lies at astronomical distance, it is modelled as an ideal on-axis point: in the optics run it is a monochromatic 511 keV far-field beam whose photons arrive parallel to the optical axis and are sampled uniformly over the incident aperture A_{inc} of Eq. (1). No finite angular size is assigned to the source itself; the angular structure that reaches the focal plane is produced entirely by the lens—the 30 arcsec mosaic spread, the XOP/CRYSTAL rocking curve, and the $f = 10 \text{ m}$ focusing geometry—so that the photons recorded at the focal plane converge to within about 0.4° of the axis and forms a focused spot rather than a single ray. Three independent seeds of 5×10^4 primaries each give 1.5×10^5 transported photons, of which 37,256 are diffracted and cross the focal plane; 37,194 of these (99.8%) fall within the $r = 1.898 \text{ cm}$ Be entrance window, with focused-spot radii $r_{90} = 1.03 \text{ cm}$ and $r_{99} = 1.25 \text{ cm}$. The within-window crossings form a MEGALib

Table 1: Background-source layers used by the current detector-coupled chain. The table defines source construction only; anticoincidence, topology selection, and the line window are applied later in Section 3.4.

Layer		Physical role	Current implementation
Prompt transport	atmospheric	Direct detector hits from atmospheric secondaries	EXPACS/PARMA full-sphere source definitions at 34° latitude, 100° longitude, 38 km altitude, $R_c = 11.6$ GV, and $W = 118.3$; γ , n , $e^\pm$, p , α , and $\mu^\pm$ over 20 equal- μ differential zenith bins including both down-going and up-going hemispheres; 25,210,216 generated primaries.
Atmospheric line	511 keV	Atmospheric positron-annihilation photons that are spectrally degenerate with the science line	Separate monoenergetic full-sphere component over 20 equal- μ bins (10 per hemisphere); day-15 $\Phi_{4\pi} = 0.0515559 \text{ ph cm}^{-2} \text{ s}^{-1}$; 3.0×10^6 generated photons, 5143.83 s effective exposure.
Activation production		Radioisotope production and production-position history	Same atmospheric field transported as a BUILDUP activation-production stream, again retaining both hemispheres and all eight particle families; 25,210,216 generated primaries.
Fixed delayed population		Radioactive activity at the adopted day-15 state after continuous exposure	Production-decay accumulation with NUBASE-checked ground-state half-lives; 141.83 Bq for the mass-complete reference and 36.4525153 Bq for the final directionally graded configuration.
Production-position decay transport		Delayed decay particles emitted from sampled production positions	Equal-activity sampled point sources from the weighted production table; the final configuration transports 10^6 delayed triggers for each of seven positive activation families, while the electron-minus family is a finite-buildup zero observation with no fictitious transport exposure.

EventList source. Each entry supplies its recorded position and direction at the Be aperture and initiates a new detector-side Monte Carlo transport through the detector/cryostat model.

The EventList-source normalization relates the detector-side transport to the reference flux. At the detector entrance the focused stream carries the rate

$$R_{fp} = F_{\text{TOA}} A_{\text{eff}}(511 \text{ keV}) T_{\text{atm},45}. \quad (11)$$

Here $A_{\text{eff}}(511 \text{ keV}) = 20.1 \text{ cm}^2$ is the on-axis Laue effective area, F_{TOA} is defined at the top of the atmosphere, and $T_{\text{atm},45} = 0.6520$ is the day-15 transmission along the fixed 45° -elevation axis; numerically this is $1.31 \times 10^{-3} \text{ s}^{-1}$ at the reference flux. Equivalently, the generating run corresponds to an observation time $N_{fp}/[F_{\text{TOA}} A_{\text{eff}} T_{\text{atm},45}]$, and each EventList entry carries the inverse of that exposure as its rate weight; in the intermediate cut-flow bookkeeping of Section 3.4 the same sample is initially normalized to a unit EventList input rate before this scale is applied. After the common detector response and event selection, the surviving signal rate is $9.86228 \times 10^{-4} \text{ s}^{-1}$. Dividing out the atmosphere gives an instrument-only selected effective area $A_{\text{sel,instr}} = S_{W_{511}}/(F_0 T_{\text{atm},45}) = 15.1254 \text{ cm}^2$; the atmosphere-folded conversion from top-of-atmosphere flux

to selected rate is $S_{W_{511}}/F_0 = 9.8623 \text{ cm}^2$. Because the optics run contains only source photons, the EventList source contains only signal photons; all backgrounds enter through the streams of Section 3.2.

The effective area quoted above is the on-axis value, representing best-case pointing. A point source offset from the optical axis, or residual attitude jitter during flight, reduces and shifts the focused response through the finite angular acceptance (field of view) of the Laue lens [11, 13]; this off-axis dependence is treated as a pointing systematic on the signal effective area rather than as a change to the on-axis reference construction. The current side-entry response is a fixed 45° -elevation reference, not a simulated low-elevation Galactic-centre observation.

3.3 Background-led geometry design and statistical treatment

The geometry design begins with the background that reaches the science window. For the mass-complete reference calculation, we group final events by incident particle family and, for delayed events, by parent isotope and production volume. The prompt map identifies the directions that require active coverage; the delayed map identifies nearby materials that feed the line window after activation. This is the same component-first logic used in detailed COSI and DIXE background studies [17, 18]. It leads to the final graded-shield geometry used for the full calculation: 40 mm BGO around the side, a 30 mm bottom disk, and a 10 mm top annulus. The side optical aperture, 3 mm Al mechanical shell, Kapton layer, detector, and cryostat are retained, and there is no external W layer. The final geometry is thus described by one physical question: where is active coverage most useful for the line-window background?

Before these stages, Cosima energy deposits are aggregated by TES pixel. Each pixel readout then receives an independent Gaussian energy response with $\Delta E_{\text{FWHM}} = 0.420 \text{ keV}$ ($\sigma = 0.17836 \text{ keV}$); measured hits below 0.3 keV are discarded, and the surviving measured hits are summed before the line window, active veto, and topology tests. The focused, prompt, and delayed catalogues use fixed response seed 26071301, while the independently convolved atmospheric-line catalogue uses fixed seed 1026071308. With Gaussian broadening disabled, the response calculation reproduces the transport-level cut-flow exactly. A 64-realization response ensemble assesses numerical stability and is used as a numerical response-integration check rather than as an additional physical systematic.

All signal and background streams are summarized at three common stages: after the energy window, after the active veto, and after the topology/field-of-view test. For stream k and stage j , the physical rate and its Monte Carlo counting error are

$$\hat{R}_{k,j} = \sum_{i \in (k,j)} w_i, \quad \sigma_{k,j} = \left(\sum_{i \in (k,j)} w_i^2 \right)^{1/2}, \quad N_{\text{eff},k,j} = \frac{\hat{R}_{k,j}^2}{\sigma_{k,j}^2}, \quad (12)$$

where w_i is the physical rate carried by record i . We also report the stage survival $q_{k,j} = \hat{R}_{k,j}/\hat{R}_{k,\text{window}}$ and the final background share $f_k = \hat{R}_{k,\text{final}}/\sum_b \hat{R}_{b,\text{final}}$. Counts and rates are both shown because the streams have different exposure weights: counts measure Monte Carlo support, whereas rates measure the physical background.

Within an independently normalized transported component the event weights are constant, so a two-sided 95% Garwood interval for the Poisson count can be multiplied by the component weight to obtain an exact rate interval. A zero selected count with a defined transport exposure is treated in the same way and therefore gives an upper limit rather than a zero-background claim. The finite- BUILDUP e^- family is different: no delayed source or transport exposure exists, so no fictitious Garwood limit is assigned. Focused-signal acceptance is binomial; its two-sided 95% interval is Clopper–Pearson. The central mission projection uses the estimated rates. A second, conditional componentwise transport-counting endpoint uses the upper Garwood endpoint for every transported background component and the lower Clopper–Pearson endpoint for the signal. Summing these component endpoints is not a joint-coverage interval and is not full 95-percent coverage; it excludes BUILDUP-yield, finite- M position sampling, zero-family, nuclide-mixture, radiation-field, cross-section, and detector systematics. These calculations used SciPy 1.8.0.

For time bin ℓ , live fraction L_ℓ , and duration Δt_ℓ , the cumulative signal and background counts are

$$N_s(T) = \sum_{\ell \leq T} S_\ell L_\ell \Delta t_\ell, \quad N_b(T) = \sum_{\ell \leq T} B_\ell L_\ell \Delta t_\ell. \quad (13)$$

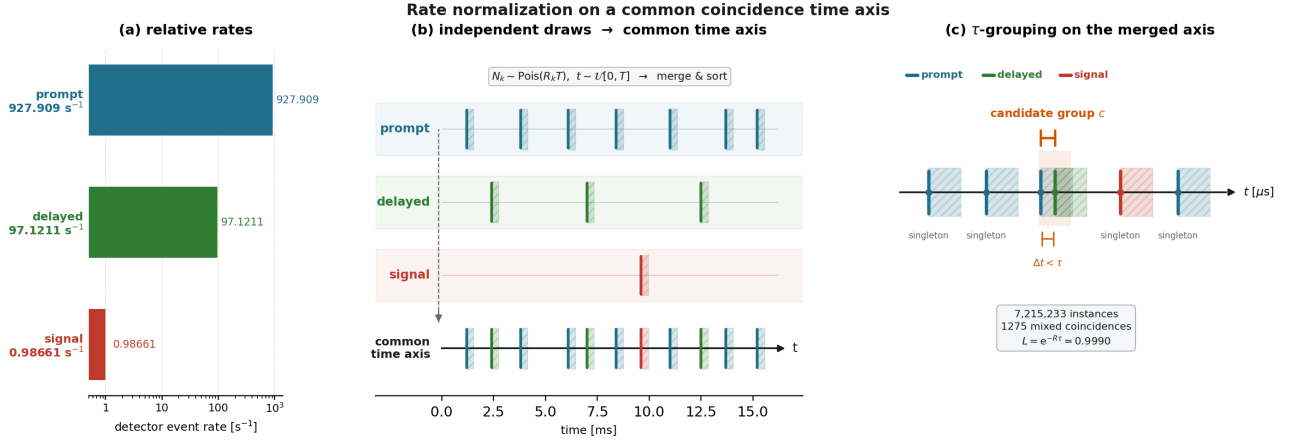


Figure 4: Reference three-stream validation of rate normalization on a common coincidence time axis. (a) Full-band detector event rates (928, 97.1, and 0.987 s^{-1}). (b) Independent Poisson draws of the three streams are merged and time-ordered; color marks stream identity and hatched bands mark each event’s temporal footprint. (c) Events with $\Delta t \leq \tau = 1 \mu\text{s}$ form one candidate group; the box gives the reference-exposure accidental-coincidence accounting. Tick positions are schematic. The final mission fold additionally includes atmospheric-line occupancy through Equation (15).

We use the transparent counting metric $Z(T) = N_s(T)/\sqrt{N_b(T)}$. Because the signal rate is linear in source flux, the corresponding diagnostic threshold is $F_{3\sigma}(T) = F_0 3/Z(T)$. The central curve and the conditional componentwise transport-counting endpoint are propagated through the same 81-bin trajectory. Their separation measures finite transported Monte Carlo support under the stated conditioning; it is not a confidence band for atmospheric fields, cross sections, or detector calibration.

3.4 Detector event selection and veto definitions

The reference coincidence calculation places the focused-signal, prompt atmospheric, and delayed-activation catalogues on one explicit time axis and applies two veto layers in sequence. The independently transported atmospheric 511 keV catalogue receives the same detector response and event-wise selection. Its full-band detector occupancy is added to the final mission-time live-factor calculation, as specified below. This distinction separates the reference three-catalogue coincidence demonstration from the final four-component rate budget.

3.4.1 Time axis and rate normalization

Each stream enters with a per-record rate weight r_{km} from the source normalization of Section 3.2, where k labels the stream and m labels a record in that stream’s event catalogue; the stream total rate is $R_k = \sum_m r_{km}$. To place three independently normalized catalogues on one physical time axis, the streams are combined by a Poisson superposition: for stream k the number of instances is drawn as

$$N_k \sim \text{Pois}(R_k T), \quad (14)$$

records are selected with probabilities r_{km}/R_k with replacement, each is assigned a uniform random time in the reference window $[0, T]$, and all streams are merged and time-ordered into a single detector time axis carrying the three physical rates. Event instances separated by no more than $\tau = 1 \mu\text{s}$ are grouped into a coincidence candidate group c . In the reference validation (Figure 4) the full-band detector rates superposed on this axis are about 928 s^{-1} (prompt), 97.1 s^{-1} (delayed activation), and 0.987 s^{-1} (focused signal), so the coincidence axis is strongly prompt-dominated; over the reference exposure the 7,215,233 event instances yield 1275 mixed-stream coincidences, giving an accidental live factor $\simeq 0.9990$ (the line window and shield threshold below then act on the resulting candidate groups). The normalized summed-energy spectrum of these three reference streams is shown in Figure 5: the window is line-dominated, with a sparsely sampled continuum.

The explicit Poisson realization in Figures 4 and 5 is a reference validation and does not contain the atmospheric-line catalogue. In the final trajectory calculation, the selected atmospheric-line rate is propagated

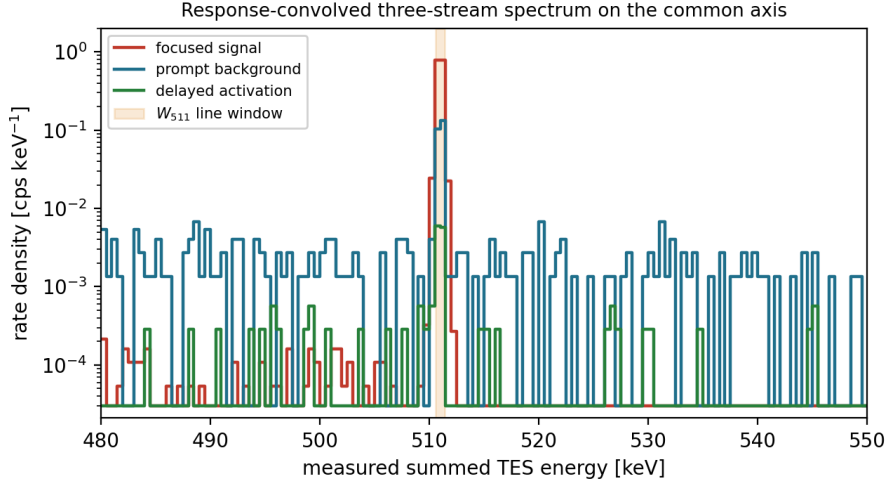


Figure 5: Response-convolved summed-energy spectra of the three reference streams on the common axis before vetoes. The focused signal, prompt, and delayed streams all concentrate at the 511 keV line while the continuum is sparsely sampled. Absolute selected line-window rates are given in Table 2.

separately and its detector occupancy is included in the live fraction for time bin ℓ :

$$L_{\ell} = \exp \left\{ -\tau \left[R_{\text{prompt},\ell}^{\text{occ}} + R_{\text{delayed},\ell}^{\text{occ}} + R_{\text{atm511},\ell}^{\text{occ}} \right] \right\}. \quad (15)$$

At the line-source day-15 reference, the atmospheric catalogue contains 1,041,505 events with at least one TES or active-shield hit, giving $R_{\text{atm511}}^{\text{occ}}(15) = 202.477 \text{ s}^{-1}$. The focused full-band rate at the reference source flux is 0.987 s^{-1} , so $R_{\text{sig}}^{\text{occ}} \tau \simeq 9.9 \times 10^{-7}$ and signal self-coincidence is neglected under the weak-source approximation. A brighter-source application would require adding $R_{\text{sig},\ell}^{\text{occ}}$ to the exponent. The active anticoincidence gate compares TES and active-shield deposits within the same τ window, and the topology reconstruction determines which hits belong to one incident photon. The same coincidence window and shield threshold are used for every transported component. Within each trajectory bin the rates are treated as stationary: the atmospheric column and activation inventory evolve on mission-day scales, whereas the coincidence calculation spans microseconds to seconds. Independent constant-rate arrivals may therefore be superposed as homogeneous Poisson processes. The mission fold uses their expectation values rather than adding a new stochastic realization in every bin; the explicit three-stream realization validates only the grouping and accidental-coincidence accounting.

3.4.2 Active anticoincidence gate

The first layer is an active anticoincidence gate on this common time axis. For each candidate group c the TES and active-shield energy sums are

$$E_{\text{TES}}^{(c)} = \sum_{j \in c} E_{\text{TES},j}, \quad E_{\text{sh}}^{(c)} = \sum_{j \in c} E_{\text{sh},j}. \quad (16)$$

The group enters the hard line window if $E_{\text{TES}}^{(c)}$ falls in

$$W_{511} = [510.58, 511.42] \text{ keV} \quad (511 \text{ keV} \pm 420 \text{ eV}), \quad (17)$$

which is the primary reference selection window, retaining most of the line response while suppressing continuum background. The group passes the active shield if

$$E_{\text{sh}}^{(c)} < E_{\text{veto}}, \quad E_{\text{veto}} = 50 \text{ keV}, \quad (18)$$

where for the baseline CsI configuration the shield sum is read from the CsI active-shield volumes in post-processing. This layer is the dominant source of rate-level background suppression. After the per-pixel detector response and threshold are applied, it reduces the prompt background from 148 to 68 events ($0.1146 \rightarrow$

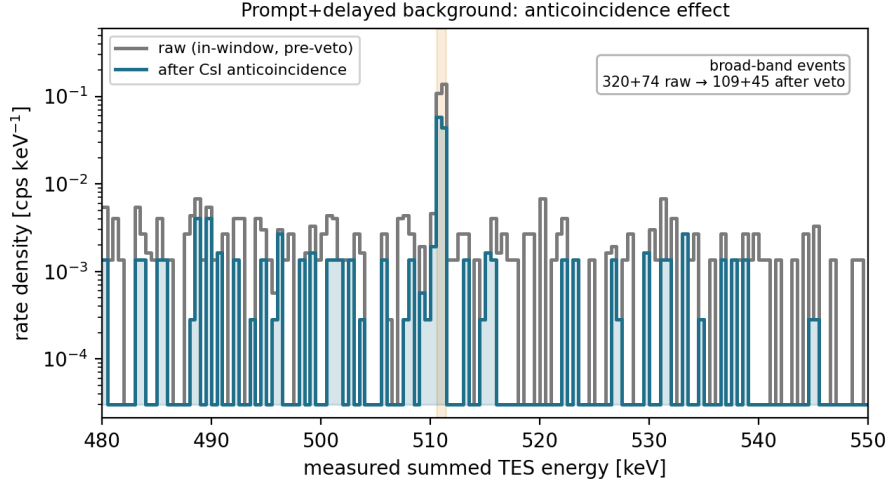


Figure 6: Effect of the CsI anticoincidence gate on the prompt+delayed background spectrum (in-window pre-veto vs after the shield gate). Shape-diagnostic re-derivation on the surviving reference-detector catalogues.

0.04612 s^{-1} , 40.24% surviving) and delayed activation from 40 to 26 events (65.00%), while retaining all 28534 focused-signal events (zero shield deposit, 100%); see Table 2. Its effect on the background spectrum is shown in Figure 6.

The quoted flux thresholds apply to a line whose intrinsic width, after detector response, remains small compared with this window; a substantially broadened source line would require convolving the source line profile with the response and re-optimizing the window.

3.4.3 Compton topology and field-of-view consistency

The second layer applies a Compton-kinematic topology and field-of-view consistency test to the TES hits that survive the active-shield gate; its physical basis is Compton-camera cone reconstruction [30]. An incident photon of energy E_0 Compton-scatters at a first interaction point $\mathbf{r}_1$, depositing E_1 , and the scattered photon of energy $E_{\text{sc}} = E_0 - E_1$ propagates to a second interaction point $\mathbf{r}_2$; the scattering angle follows from

$$\cos \theta_C = 1 - m_e c^2 \left(\frac{1}{E_{\text{sc}}} - \frac{1}{E_1 + E_{\text{sc}}} \right). \quad (19)$$

Given the scattered-photon direction $\hat{\mathbf{u}}_{12} = (\mathbf{r}_2 - \mathbf{r}_1)/|\mathbf{r}_2 - \mathbf{r}_1|$, the source direction is constrained to a reconstruction cone with apex $\mathbf{r}_1$, axis $-\hat{\mathbf{u}}_{12}$, and half-angle θ_C , whose projection on the sky is the event circle. We use this first-scatter cone as an aperture-consistency test: the event is retained when the cone intersects the side Be-window disk (radius 1.898 cm), i.e. whether the reconstructed arrival direction is consistent with entry through the aperture.

Because the interaction position within a single TES pixel ($1.5 \times 1.5 \times 3.0 \text{ mm}$) is unresolved, each hit is represented not by its centroid alone but by $8 + 1 = 9$ representative points—the eight pixel corners and the centroid—carrying the sub-pixel position uncertainty explicitly into the geometric reconstruction. A two-hit event in a given scatter order therefore enumerates $9 \times 9 = 81$ first/second position combinations and constructs one reconstruction cone per combination (each cone sampled at 24 azimuths, propagated to the Be-window plane, and tested against the disk by both its sampled points and the connecting segments); the order is deemed field-of-view consistent if any of these 81 trajectories yields a cone that intersects the disk. This is a deliberately permissive consistency test: an event is retained if any sub-pixel interaction position is consistent with entry through the aperture. The layer consequently retains about 95% of the in-window multi-hit events and removes only about 10–14% of the background, consistent with its role as a loose consistency veto rather than an imaging reconstruction.

The three event classes are handled accordingly. Single-pixel events have no scatter topology and are retained as calorimetric line candidates. A two-hit event provides two positions and energies but no internal information

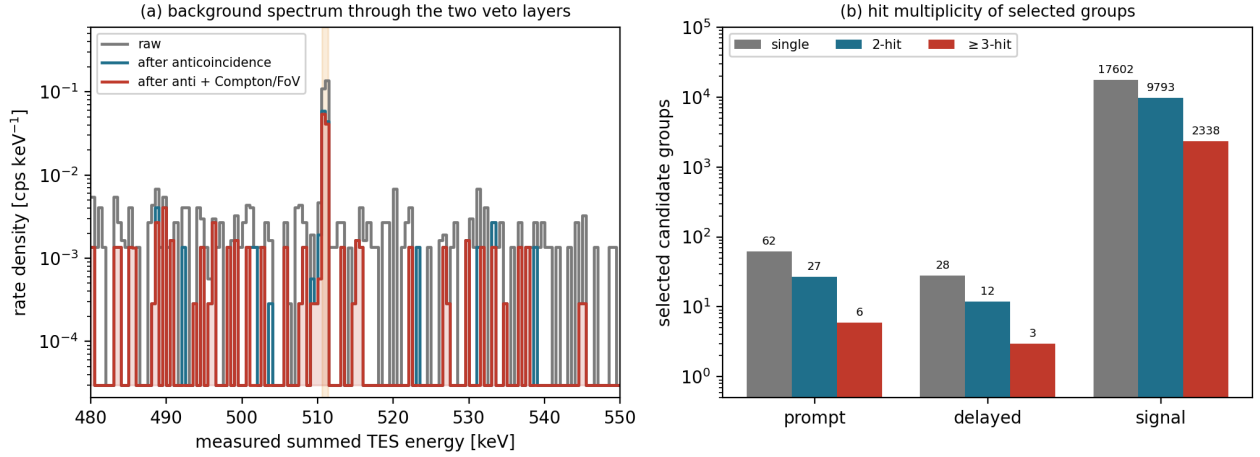


Figure 7: (a) Response-convolved background spectrum through the two veto layers (raw; after anticoincidence; after anticoincidence+Compton/FoV). (b) Hit multiplicity of the selected candidate groups per stream in the 480–550 keV band. Absolute line-window rates are given in Table 2.

to order the interactions, so both scatter orders are enumerated over the 81 trajectories above, and the event is retained if either order is field-of-view consistent and rejected only if neither order admits any kinematically valid cone. Three-hit and higher-multiplicity events contain internal scatter vertices and can be ordered from geometric–kinematic consistency: at each internal vertex the scattering angle can be obtained both from the deposited energies through the Compton formula and from the angle between successive flight directions, and the two should agree for the correct order. All orders are enumerated (up to six TES hits; higher multiplicities are retained without ordering), the best order minimizes the Compton sequence residual

$$Q = \sum_{i=1}^{n-2} (\cos \theta_{C,i} - \hat{\mathbf{u}}_{i-1,i} \cdot \hat{\mathbf{u}}_{i,i+1})^2 \quad (20)$$

(ties broken by the longest first-scatter lever arm), and the same 81-trajectory cone–disk test is applied to its first two points. Events with no valid order are classified as unreconstructed. In the response-convolved baseline, events that pass the active shield are sorted by the topology test into single-site, field-of-view-consistent reconstructed multi-hit, and vetoed classes: the prompt stream comprises 42 single-site, 21 reconstructed multi-hit, and 5 vetoed events (68 total); the delayed stream 19, 7, and 0 (26 total); and the focused signal 17267 single-site, 10726 reconstructed multi-hit, and 541 vetoed events (28534 total), giving 63, 26, and 27993 selected events, respectively. In the 480–550 keV broad band, the same response and selection resolve the selected focused-signal groups into 17602 single-site, 9793 two-hit, and 2338 ≥ 3 -hit events (with 370 two-hit and 262 ≥ 3 -hit events Compton-vetoed); the corresponding selected multiplicities are 62/27/6 for prompt and 28/12/3 for delayed events (Figure 7).

Unreconstructed events remain in the baseline rate accounting but are not labelled as field-of-view-passing candidates. The role of multi-hit acceptance can be quantified directly: a single-site-only selection would retain 61.68% of the focused-signal rate and 67.17% of the selected background rate in W_{511} , reducing the simple $S/\sqrt{B}$ figure to 0.753 of the baseline value. The adopted permissive multi-hit policy therefore preserves statistical reach while using topology as a consistency veto rather than as an imaging cut.

Table 2 summarizes the reference-geometry event flow with the same TES energy window, CsI veto, and Compton/field-of-view test. The first stage is the line-window sample before either veto, and the uncertainties are the quadrature sums of the event-rate weights.

An independent topology benchmark uses the MEGAlib *revan* Compton event reconstructor on 37,065 focused-signal events from a retained legacy geometry [30]. Its energy-window population is not mixed with the response-convolved reference-detector rate budget above; the benchmark is used for interaction ordering and angular-resolution diagnostics. For two-hit ordering it gives two quantitative results. First, the fraction of two-hit events with at least one kinematically valid order is 100.0% (10,421/10,421), as expected analytically—for

Table 2: Reference-geometry event-level W_{511} cut-flow. Focused-signal rates are normalized to a unit EventList input rate; the physical point-source rate is scaled separately.

Stream	Stage	Events	Rate / s^{-1}	Statistical uncertainty / s^{-1}
Prompt background	Line-window pre-veto	148	1.14624×10^{-1}	1.24549×10^{-2}
Prompt background	After CsI veto	68	4.61248×10^{-2}	5.59345×10^{-3}
Prompt background	Topology/FOV selected	63	4.27321×10^{-2}	5.38374×10^{-3}
Delayed activation	Line-window pre-veto	40	5.69229×10^{-3}	9.00030×10^{-4}
Delayed activation	After CsI veto	26	3.69999×10^{-3}	7.25627×10^{-4}
Delayed activation	Topology/FOV selected	26	3.69999×10^{-3}	7.25627×10^{-4}
Focused signal	Line-window pre-veto	28534	7.67167×10^{-1}	4.54160×10^{-3}
Focused signal	After CsI veto	28534	7.67167×10^{-1}	4.54160×10^{-3}
Focused signal	Topology/FOV selected	27993	7.52621×10^{-1}	4.49834×10^{-3}

The focused-signal rows report transfer rates for a unit EventList input rate and show the survival fraction at each selection stage; the point-source signal rate in Table 7 is scaled separately by the Laue effective area, reference flux, and atmospheric transmission.

a fully absorbed 511 keV two-hit event the scattered-photon energy cannot fall below the $E_0/3$ Compton edge, so at least one order is always valid—confirming the premise for retaining two-hit events. Second, the best order chosen by the Klein–Nishina cross section with a kinematic quality factor agrees with the order that points more nearly toward the source only 78.4% of the time, so committing to a single estimated order (as an imaging chain would) would use the wrong Compton cone for about 22% of two-hit events, whereas enumerating both orders and accepting either field-of-view-consistent one avoids this misassignment. The median $|\text{ARM}|$ of in-window Compton events relative to the true beam direction is about 24° (two-hit 23.6° ; three-hit and higher 26.3° ; only about 32% have $|\text{ARM}| < 10^\circ$), driven by the millimetre-scale scatter lever arms and pixel quantization; this explains why the topology/field-of-view layer removes only about 10–14% of the background and cannot be sharpened into an imaging-level discriminant. Finally, the fraction of unorderable events is very small (61 of 37,065, i.e. 0.16%, in revan; one in the analytic classifier), so the unreconstructed class is a small boundary population.

3.5 Mission-time trajectory and analytic rate folding

MeV balloon observations are background dominated: atmospheric secondaries and cosmic-ray-induced activation set the rate floor, and both respond to the flight environment [17, 36]. Detector-coupled Monte Carlo through a fixed mass model already provides the selected day-15 signal and background rates under a common veto and energy window (Sections 3.2–3.4). Those rates, however, are defined at a single longitude, latitude, and altitude. On a multi-day float the residual atmospheric depth, geomagnetic cutoff, and activation inventory evolve, so a constant-rate extrapolation of the day-15 snapshot is not a substitute for a trajectory fold. As in other balloon MeV programmes (for example COSI’s use of PARMA/EXPACS fields and time-dependent light curves along the flight [17]), we therefore introduce an analytic mission-time trajectory curve that maps the reference rates onto a continuous 20 d performance estimate, and we test the fold with direct Cosima transports at selected trajectory states rather than re-running full statistics at every time bin.

The trajectory used here is a *synthetic reference profile*, not balloon telemetry and not a source-visibility schedule. It comprises 81 bins from day 0 to day 20 (0.25 d spacing) with a mild mid-latitude float envelope: altitude $h = 38.75 \pm 2.5$ km, latitude $34^\circ \pm 0.25^\circ$, and longitude $100^\circ \pm 0.25^\circ$, all phase-referenced to the day-15 anchor. Altitude is retained because zero-pressure-class and related long-duration balloons experience float-altitude excursions that change residual atmospheric depth by several g cm^{-2} ; depth controls both the atmospheric particle spectra that drive prompt background and activation, and the 511 keV atmospheric transmission that multiplies the focused science rate. Latitude and longitude are retained only as a *reference* geomagnetic/geographic modulation of cutoff rigidity on that same synthetic track; they do not encode Earth occultation, target elevation limits, or repointing losses. The resulting fold therefore supports a reference-exposure statistical sensitivity under a controlled trajectory envelope, not a flight forecast for a named launch campaign.

The analytic fold is defined by six physically separate layers. Atmospheric transmission multiplies only the celestial signal and does not rescale either the local particle field that drives prompt background and activation

or the atmospheric-line source defined in Equation (3).

(1) *Signal*. For a monochromatic point source with top-of-atmosphere flux F_{511} ,

$$R_{\text{sig}}(t) = F_{511} A_{\text{eff}} T_{\text{atm},511}(t) \varepsilon_{\text{det}}, \quad (21)$$

where A_{eff} is the on-axis Laue effective area, $T_{\text{atm},511}(t)$ is the 511 keV atmospheric transmission at the trajectory state, and ε_{det} collects detector and selection efficiency at the reference response. Relative to the day-15 anchor this is equivalent to $S_{\text{sig}}(t) = S_{15} T_{\text{atm},511}(t)/T_{15,45}$ with $T_{15,45} = 0.6520$. In every bin the fixed 45° elevation (45° zenith angle) gives $T_{\text{atm},511}(t) = \exp[-\mu_{\text{air}} X_{\text{vertical}}(t)/\cos 45^\circ]$, with $\mu_{\text{air}} = 0.08736 \text{ cm}^2 \text{ g}^{-1}$. This uses the slant column along the instrument axis and remains a fixed-axis reference rather than a source-visibility calculation.

(2) *Prompt background*. At particle-family resolution the selected prompt rate is the source–response sum

$$R_{\text{prompt},j}(t) = \sum_b \Phi_b(t) K_{j,b}, \quad (22)$$

where b runs over atmospheric particle families (protons, neutrons, $e^\pm$, γ , $\mu^\pm$, ...), $\Phi_b(t)$ is the PARMA/EXPACS flux recalculated at the trajectory state, and $K_{j,b}$ is a detector response coefficient fixed once from the day-15 reference transport for selection j . Relative to the reference state this reweights family contributions by $\Phi_b(t)/\Phi_b(\text{REF})$. Within a line window the same structure may be refined to $K_{b,E,\theta}$ when energy–angle response matrices are available.

(3) *Atmospheric 511 keV line*. The separately transported day-15 line response is scaled by the full-sphere line-integrated flux:

$$R_{\text{atm}511,j}(t) = R_{\text{atm}511,j}(15) \frac{\Phi_{4\pi}(t)}{\Phi_{4\pi}(15)}, \quad R_{\text{atm}511}^{\text{occ}}(t) = R_{\text{atm}511}^{\text{occ}}(15) \frac{\Phi_{4\pi}(t)}{\Phi_{4\pi}(15)}. \quad (23)$$

Here $\Phi_{4\pi}(t)$ is evaluated from Equation (3) along the atmospheric-line proxy depth and rigidity curve, whose day-15 rigidity is 11.0 GV. This curve is distinct from the state-specific PARMA prompt field, whose day-15 rigidity is 11.6 GV. The total line flux is varied while the transported day-15 angular response is held fixed; the line transport is not rerun in every time bin.

(4) *Activation production*. For each incident family a and radioactive source-parent nuclide k in the final all-eight-family delayed inventory,

$$P_{a,k}(t) = \Phi_a(t) Y_{a,k}, \quad (24)$$

where $Y_{a,k}$ is the family-resolved yield fixed by the corresponding day-15 inventory and $\Phi_a(t)$ is the PARMA flux recalculated for that family at each trajectory state. The 141.83 Bq all-family inventory of the mass-complete reference geometry is used for the background-origin analysis, not substituted into this final-geometry fold; the fold uses the final directionally graded configuration's 36.4525153 Bq inventory.

(5) *Delayed inventory*. With decay constant λ_k , each family–parent inventory is advanced by

$$N_{a,k}(t + \Delta t) = N_{a,k}(t) e^{-\lambda_k \Delta t} + \frac{P_{a,k}(t)}{\lambda_k} (1 - e^{-\lambda_k \Delta t}), \quad A_{a,k}(t) = \lambda_k N_{a,k}(t). \quad (25)$$

(6) *Delayed selected rate*.

$$R_{\text{delayed},j}(t) = \sum_a \sum_k A_{a,k}(t) \varepsilon_{j,a,k}, \quad (26)$$

where $\varepsilon_{j,a,k}$ is the selected-rate response of source-parent nuclide k produced by family a under selection j . A SIM IA INIT daughter identity is retained as lineage evidence but does not replace the radioactive source parent in this sum. Equations (21)–(26) keep celestial transmission, prompt-particle modulation, atmospheric-line scaling, activation production, and radioactive decay as separate physical steps.

To test the analytic predictions we select and hold fixed four trajectory states—the day-15 reference (REF), a low-altitude early point (L1), a high-altitude point (H1), and a low-altitude late point (L2)—and recalculate

PARMA spectra and Cosima atmospheric sources at those coordinates without using the analytic scale as a free parameter. For a predicted scale S^{pred} from Equation (22) we report

$$Q = \frac{R^{\text{MC}}/R_{\text{REF}}^{\text{MC}}}{S^{\text{pred}}}. \quad (27)$$

Ideal agreement gives $Q \simeq 1$. The final implementation applies separate state-specific PARMA flux ratios to the day-15 response coefficient of each of the eight prompt particle families and advances each retained delayed family–parent nuclide with its matching family flux ratio. Direct trajectory transports validate the combined e^+ , n , and γ modulation in the broad TES 480–550 keV band within 2σ (maximum $|z| \simeq 0.51$ across the off-reference points); for the high-statistics e^+ band alone, Q follows the recalculated PARMA e^+ flux scale to within about 1%. These multi-point results directly validate the tested families, while the other prompt-family modulation remains an analytic source-field reweighting; all final 511 keV-window response coefficients are supplied by the day-15 detector chain.

With the final-geometry day-15 rates fixed by the full detector chain, these trajectory scales provide $S(t)$, $B(t)$, and the cumulative counts in Equation (13). Section 4 presents the resulting central curve and conditional componentwise transport-counting endpoint together, so the reader can see both the expected mission performance and the effect of limited Monte Carlo support on the same time axis.

4 Results

4.1 What sets the 511 keV line-window background?

The mass-complete reference calculation tells us which interactions must be controlled before a shield is specified. In the 510.58–511.42 keV window, the common response, active-veto, and topology/field-of-view selection leaves 63 prompt and 26 delayed Monte Carlo records. Because the streams represent different physical exposures, their weights differ; the selected records correspond to a prompt-plus-delayed diagnostic rate of $(4.64320 \pm 0.54324) \times 10^{-2} \text{ s}^{-1}$. Table 3 gives both the record counts and the weighted rates. The separately transported atmospheric 511 keV line is not included in this reference-geometry diagnostic.

Figure 8 makes the prompt result easy to read. Incident positrons contribute $2.917 \times 10^{-2} \text{ s}^{-1}$ and neutrons contribute $1.221 \times 10^{-2} \text{ s}^{-1}$. Together they supply 96.84% of the prompt rate and 89.13% of the prompt-plus-delayed diagnostic budget. The atmospheric 511 keV line is carried as the separate component defined in Section 3.2.2; under the adopted detector response and line window it is not spectrally distinguishable from the science line. All eight prompt families remain in the final transport; the decomposition is used to understand the physics, not to remove source classes from the calculation.

The delayed events show a different source-level and detector-selected distribution. Of the corrected 141.83 Bq day-15 inventory, neutron-induced products contribute 94.25% and μ^- -induced products 5.38%. The source activity is distributed mainly among CsI (63.08%), outer mechanical structures (15.71%), and cold plates (9.43%); the neutron and μ^- material distributions have $D_{\text{TV}} = 0.823$. After detector response and selection, 24 of the 26 line-window records are ^{64}Cu decays and the other two are ^{61}Cu decays, all in cold copper structures near the TES. The source inventory and selected line-window background therefore identify different material priorities: active coverage controls the dominant prompt and atmospheric paths, while copper close to the focal plane supplies the local delayed component.

The final directionally graded configuration uses a separate all-eight-family activation calculation. Its NUBASE-corrected day-15 activity is 36.4525153 Bq: neutron, μ^- , proton, and α production contribute 83.83%, 10.82%, 4.64%, and 0.69%, respectively, with smaller positive e^+ , γ , and μ^+ inventories; e^- is the finite-BUILDUP zero observation. The 78 final delayed records are carried by 60 neutron, 12 proton, five μ^- , and one α event. Their time fold is resolved jointly by incident family and exact-source parent nuclide. In particular, the one SIM lineage that starts as ^{182}W is assigned to its position-matched radioactive ^{182}Ta source parent rather than treating stable ground-state ^{182}W as an activity.

Table 3: Prompt-plus-delayed primary-window diagnostic for the reference detector; the separately transported atmospheric 511 keV line is not included. Relative uncertainties are Monte Carlo counting standard errors. Counts from independently normalized components are not directly proportional to their rate contributions.

Component	Selected records	Rate / s^{-1}	MC σ / s^{-1}	Relative MC σ / %	Total rate / %	Selected-event origin
Prompt e^+	43	2.91730×10^{-2}	4.44885×10^{-3}	15.2	62.83	incident-family tag
Prompt n	18	1.22099×10^{-2}	2.87791×10^{-3}	23.6	26.30	incident-family tag
Prompt μ^+	2	1.34910×10^{-3}	9.53958×10^{-4}	70.7	2.91	incident-family tag
Prompt subtotal	63	4.27321×10^{-2}	5.38374×10^{-3}	12.6	92.03	eight-family transport
Delayed activation	26	3.69999×10^{-3}	7.25627×10^{-4}	19.6	7.97	24 ^{64}Cu , 2 ^{61}Cu
Total	89	4.64320×10^{-2}	5.43242×10^{-3}	11.7	100.00	prompt plus delayed

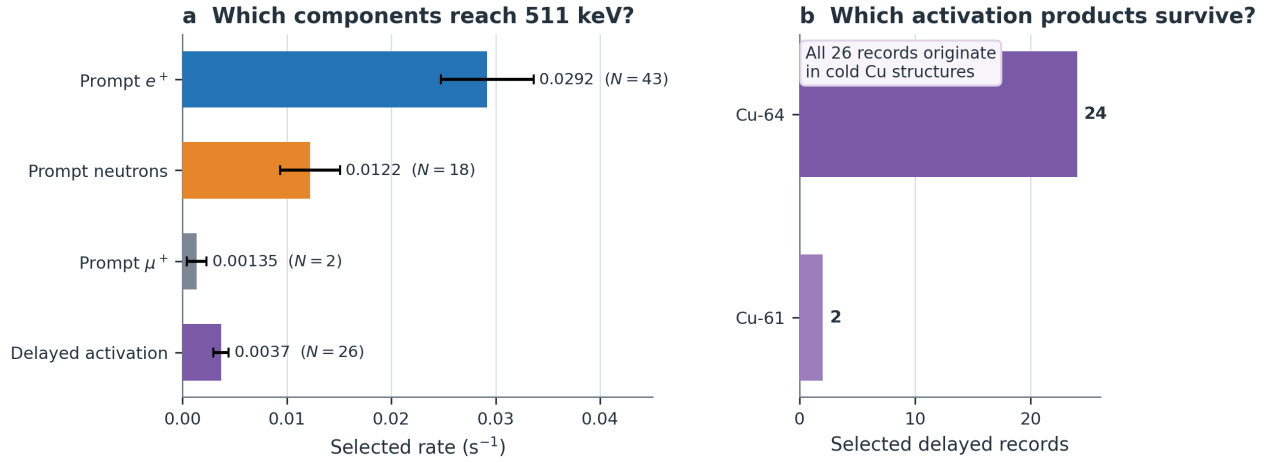


Figure 8: Physical origin of the selected prompt-plus-delayed diagnostic background in the mass-complete reference geometry; the atmospheric 511 keV line is not included. (a) Weighted rate by prompt particle family and delayed activation; error bars are Monte Carlo counting standard errors from Equation (12). (b) Parent isotope of the 26 selected delayed records. Positrons and neutrons provide 89.13% of the total selected rate, while the delayed records all originate in cold copper structures.

4.2 Final directionally graded active-shield geometry

The final active shield converts the background map into a simple geometry. The side is the most important active surface because it surrounds the long cryostat wall and the side optical opening, so it retains 40 mm BGO coverage. The bottom uses a 30 mm BGO disk, while the top uses a 10 mm annulus around the service opening. The focused beam still enters through the aligned side aperture. A thin Al/Kapton outer enclosure remains, with no continuous external W layer. Figure 9 connects this layout to the event selection and final background budget; Table 4 lists the dimensions. These values define the only optimized geometry used in the following full-chain and mission calculations.

The resulting shield is directional rather than uniform: it places the deepest active layer around the side and uses smaller end layers where the geometry has a shorter exposed path. This design also keeps the optical route easy to read: the collimator remains a local element in the beam path, whereas the outer enclosure does not add a second continuous high- Z layer. The geometry drawing, cut-flow, and component budget are therefore three linked views of the same final design.

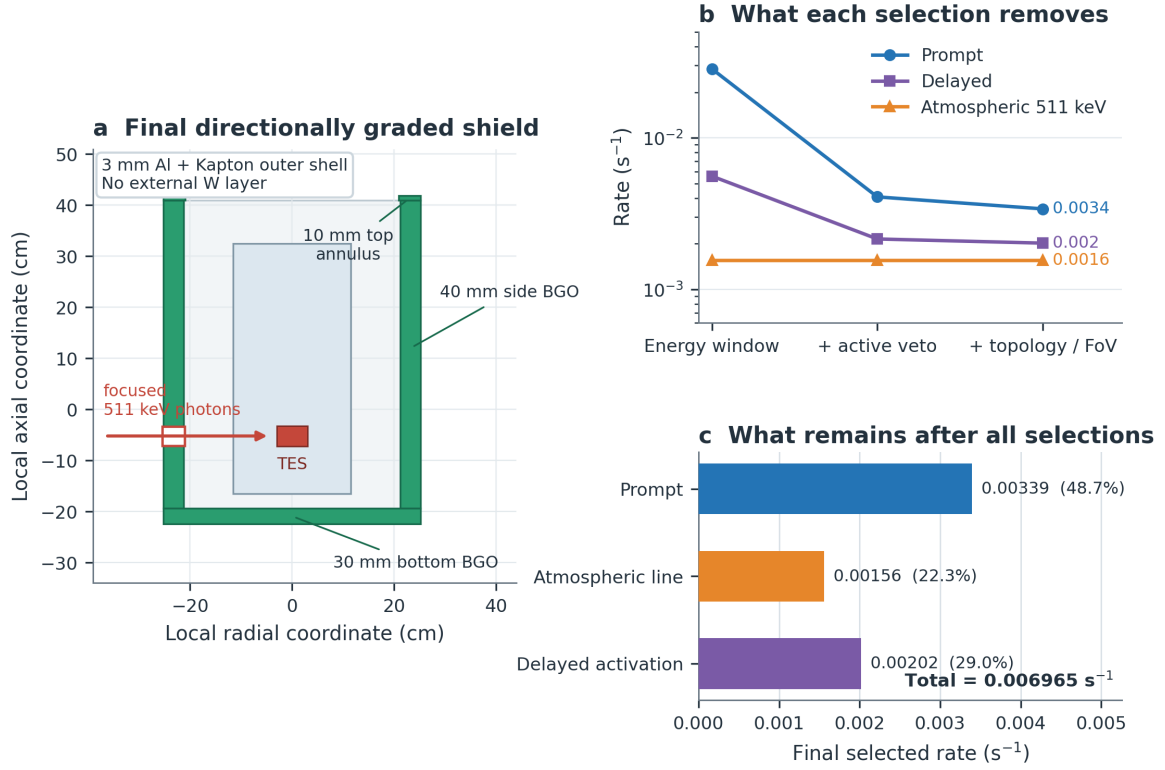


Figure 9: From the final graded-shield geometry to the selected background. (a) Cross-section of the directionally graded BGO shield; dimensions are read from the generated geometry manifest. The cryostat and TES symbols are simplified only to expose the shield layout and side beam path. (b) Prompt, delayed, and atmospheric-line rates after the energy window, active veto, and topology/field-of-view selection. (c) Composition of the final, response-convolved $6.96486 \times 10^{-3} s^{-1}$ background.

Table 4: Final directionally graded active-shield geometry. Radial and axial coordinates are local to the detector–cryostat frame.

Element	Final setting	Physical role
Side BGO	40 mm; $r = 21.2\text{--}25.2$ cm, $z = -19.4\text{--}40.9$ cm	Continuous active coverage around the long side wall while preserving the optical cut
Bottom BGO	30 mm; $r = 0\text{--}25.2$ cm, $z = -22.4\text{--}19.4$ cm	Active coverage of the lower end surface
Top BGO	10 mm annulus; $r = 20.9\text{--}25.2$ cm, $z = 40.9\text{--}41.9$ cm	Covers the outer top edge while leaving the central service opening
Side optical path	Existing Boolean aperture and aligned Al/Be windows plus W collimator	Carries the focused 511 keV beam to the TES array
Outer enclosure	3 mm Al plus Kapton; no continuous external W layer	Mechanical envelope and electrical insulation
BGO offline anticoincidence	$E_{sh}^{(c)} < 50$ keV	Event-by-event sum of recorded energy deposits in the active BGO volumes

4.3 Full-chain background and statistical support

Table 5 follows each final-geometry stream through the same event sequence shown in Figure 9b. “Pre-veto” means that the TES energy already lies in the primary line window. Record counts show the Monte Carlo support at each step; final rates give the physical result after the different source exposures have been applied.

The active shield does most of the prompt rejection. It lowers the response-convolved line-window prompt rate from 2.84914×10^{-2} to $4.07188 \times 10^{-3} s^{-1}$, so only 14.29% survives that stage. The topology/field-of-view test then lowers it to $3.39303 \times 10^{-3} s^{-1}$. For delayed activation, the corresponding rates are 5.58116×10^{-3} , 2.14813×10^{-3} , and $2.01657 \times 10^{-3} s^{-1}$; because the delayed families have different event weights, the

Table 5: Response-convolved primary-window cut-flow for the final graded-shield geometry at the day-15 reference state. The signal rate is scaled to $F_0 = 10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$; background fractions use the total final background rate.

Stream	Pre-veto records	Active-veto pass	Final records	Active survival / %	Topology survival / %	Final rate / s^{-1}	Background / %
Eight-family prompt	42	6	5	14.29	83.33	3.39303×10^{-3}	48.72
All-eight-family delayed	346	83	78	38.49	93.88	2.01657×10^{-3}	28.95
Atmospheric 511 keV line	8	8	8	100.00	100.00	1.55526×10^{-3}	22.33
Focused signal	28566	28566	28010	100.00	98.05	9.86228×10^{-4}	–
Total background	396	97	91	–	–	6.96486×10^{-3}	100.00

active and topology survivals in Table 5 are rate ratios (38.49% and 93.88%), not record-count ratios. Before detector-response convolution, nine transported atmospheric-line records lay in the primary window. After the independent 420 eV FWHM per-pixel response and the 0.3 keV measured-hit threshold were applied, eight remained in the measured window. No atmospheric-line pixel hit was removed by the threshold in this response realization; one response-convolved event instead moved below the lower window boundary. All eight measured-window records pass the active veto and topology test, leaving the rate unchanged at $1.55526 \times 10^{-3} \text{ s}^{-1}$ through those two selection stages; under the adopted energy and topology observables, a 511 keV photon entering through the open path is not distinguishable from a source photon. The focused signal behaves differently: all 28566 line-window records pass the active veto, and 28010 (98.05%) pass the final topology test.

At the day-15 reference state, after all selections, prompt particles contribute 48.72% of the background, the atmospheric line 22.33%, and delayed activation 28.95%. Thus the active shield removes most prompt and delayed events accompanied by shield energy without rejecting the focused beam, while the remaining budget naturally becomes a mixture of prompt leakage, a line-like atmospheric term, and local activation.

The five final prompt records comprise two e^+ and three neutron records. The other six prompt families have zero selected records in the available sample. Table 6 retains those zeros and their different event weights, because zero selected records give a rate upper limit. The photon family has the largest event weight and therefore sets most of the conditional componentwise prompt endpoint.

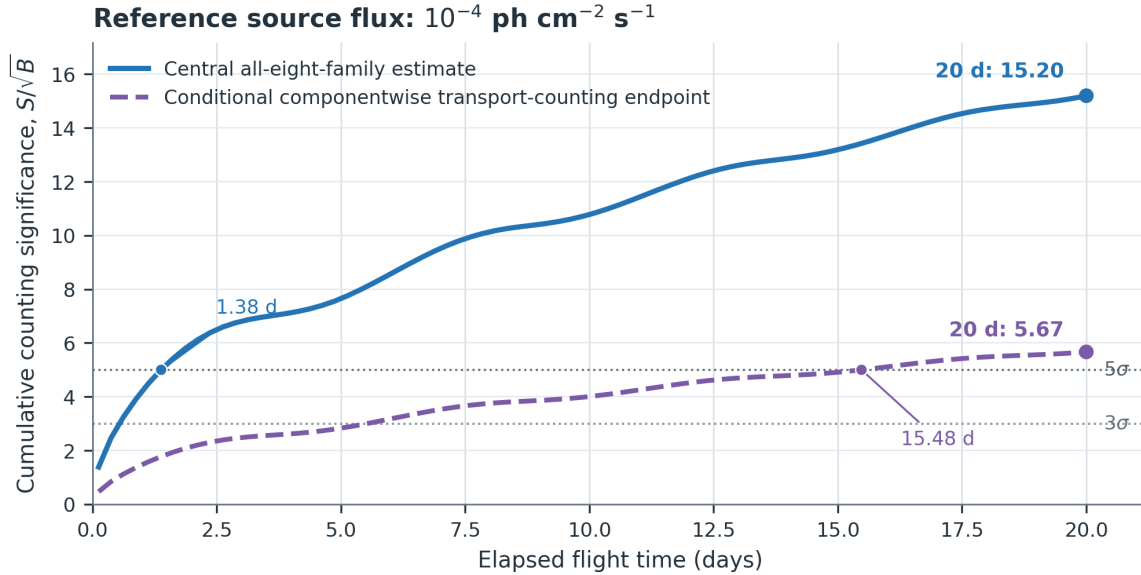
Table 6: Transport-counting precision of the optimized day-15 final background. For components with defined transport exposure, intervals are two-sided 95% Garwood intervals scaled by the listed event weight. Subtotal and total upper values are componentwise endpoint sums, not a joint-coverage interval or detector systematic uncertainty. The finite-BUILDUP e^- zero has no fictitious exposure or Garwood endpoint.

Component	Final records	Weight / s^{-1}	Central rate / s^{-1}	Exact 95% rate interval / s^{-1}	Total central / %
Prompt α	0	6.78971×10^{-4}	0	$[0, 2.50464 \times 10^{-3}]$	0
Prompt e^-	0	6.78761×10^{-4}	0	$[0, 2.50387 \times 10^{-3}]$	0
Prompt e^+	2	6.78847×10^{-4}	1.35769×10^{-3}	$[1.64423 \times 10^{-4}, 4.90446 \times 10^{-3}]$	19.49
Prompt γ	0	5.42680×10^{-3}	0	$[0, 2.00188 \times 10^{-2}]$	0
Prompt μ^-	0	6.77164×10^{-4}	0	$[0, 2.49798 \times 10^{-3}]$	0
Prompt μ^+	0	6.77705×10^{-4}	0	$[0, 2.49997 \times 10^{-3}]$	0
Prompt n	3	6.78446×10^{-4}	2.03534×10^{-3}	$[4.19736 \times 10^{-4}, 5.94812 \times 10^{-3}]$	29.22
Prompt p	0	6.77234×10^{-4}	0	$[0, 2.49824 \times 10^{-3}]$	0
Prompt subtotal	5	–	3.39303×10^{-3}	upper sum: 4.33761×10^{-2}	48.72
Delayed α	1	3.48183×10^{-7}	3.48183×10^{-7}	$[8.81522 \times 10^{-9}, 1.93995 \times 10^{-6}]$	0.005
Delayed e^-	0	–	0	finite-BUILDUP zero; no interval	0
Delayed e^+	0	2.15006×10^{-9}	0	$[0, 7.93131 \times 10^{-9}]$	0
Delayed γ	0	5.46390×10^{-9}	0	$[0, 2.01557 \times 10^{-8}]$	0
Delayed μ^-	5	4.20909×10^{-6}	2.10454×10^{-5}	$[6.83339 \times 10^{-6}, 4.91130 \times 10^{-5}]$	0.30
Delayed μ^+	0	7.94287×10^{-10}	0	$[0, 2.93003 \times 10^{-9}]$	0
Delayed n	60	3.28039×10^{-5}	1.96823×10^{-3}	$[1.50197 \times 10^{-3}, 2.53351 \times 10^{-3}]$	28.26
Delayed p	12	2.24533×10^{-6}	2.69439×10^{-5}	$[1.39223 \times 10^{-5}, 4.70656 \times 10^{-5}]$	0.39
Delayed subtotal	78	–	2.01657×10^{-3}	upper sum: 2.63166×10^{-3}	28.95
Atmospheric 511 keV line	8	1.94408×10^{-4}	1.55526×10^{-3}	$[6.71451 \times 10^{-4}, 3.06448 \times 10^{-3}]$	22.33
Total background	91	–	6.96486×10^{-3}	componentwise upper endpoint: 4.90722×10^{-2}	100.00

The 64 deterministic response seeds provide a separate numerical-integration check. Their day-15 background has mean $6.94643 \times 10^{-3} \text{ s}^{-1}$ and sample standard deviation $3.89046 \times 10^{-4} \text{ s}^{-1}$; the prespecified primary-seed value $6.96486 \times 10^{-3} \text{ s}^{-1}$ differs from that mean by 0.047 sample standard deviations. We therefore use the fixed-seed value for the primary result and keep the response-seed ensemble distinct from the conditional componentwise transport-counting endpoint in Table 6.

4.4 Mission performance of the final geometry

The final rates were folded through all 81 trajectory bins. The accidental-coincidence live factor remains between 0.97165 and 0.97985. At 20 d, the central curve contains 1650.29 focused-signal counts and 11789.08 background counts, giving $Z = 15.199$. Figure 10 shows how the significance accumulates rather than reporting only the endpoint.



Conditional endpoint: componentwise transported-count upper endpoints and signal lower bound; not a full 95% coverage interval.

Figure 10: Cumulative counting significance of the final graded-shield geometry for a continuously observed $10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$ point source along the 20 d reference trajectory. The solid curve uses the central selected rates. The dashed curve uses the upper exact 95% counting endpoints for every background component and the lower exact 95% signal-acceptance endpoint. Their componentwise combination is conditional, not a joint 95% interval. The curves reach 5σ at 1.38 and 15.48 d, respectively, and end at 15.20 and 5.67.

Table 7: Final graded-shield mission projection. The endpoint column propagates componentwise Garwood background upper endpoints and the Clopper–Pearson signal lower endpoint through the same trajectory. It is conditional on the transported samples and selected family–nuclide mixture and is not a joint-coverage interval.

Quantity	Central estimate	Conditional componentwise transport-counting endpoint
Day-15 background / s^{-1}	6.96486×10^{-3}	4.90722×10^{-2}
Day-15 signal at F_0 / s^{-1}	9.86228×10^{-4}	9.80445×10^{-4}
$T_{3\sigma}$ / d	0.545	5.519
$T_{5\sigma}$ / d	1.382	15.476
$Z_{20\text{d}}$	15.199	5.667
$F_{3\sigma}(20 \text{ d})$ / $\text{ph cm}^{-2} \text{ s}^{-1}$	1.9738×10^{-5}	5.2934×10^{-5}

The two columns answer two useful questions. The central estimate is the expected result from the simulated rates: it reaches 3σ in 0.545 d, 5σ in 1.382 d, and a 20 d flux threshold of $1.9738 \times 10^{-5} \text{ ph cm}^{-2} \text{ s}^{-1}$. The conditional endpoint asks how the answer changes if every transported background component is moved to its exact upper counting endpoint and the signal acceptance to its lower endpoint. It reaches 5σ in 15.476 d and gives a 20 d threshold of $5.2934 \times 10^{-5} \text{ ph cm}^{-2} \text{ s}^{-1}$. This endpoint does not have 95% joint coverage: the componentwise sum has no joint-coverage guarantee and omits finite-BUILDUP zero-family, finite- M source sampling, and nuclide-mixture uncertainty. The large separation between the curves is driven mainly by the zero-count, high-weight prompt-photon component in Table 6; it shows exactly where additional Monte Carlo statistics would most improve the precision of the prediction.

5 Discussion

The main result is a connected physical explanation of the final design. In the reference geometry, positrons and neutrons dominate the selected prompt rate, while the selected activation events come from copper close to the TES rather than from the volumes with the largest total activity. The final graded-shield geometry responds to the first pattern with deep side BGO coverage and treats the second as a local activation term in the same detector selection. At the day-15 reference state, after the common veto and topology test, the background is $6.965 \times 10^{-3} \text{ s}^{-1}$ and the focused signal retains 98.05% of its line-window records. The geometry, background budget, and mission curve are therefore three views of the same result.

This analysis sequence is consistent with methods used in recent instrument-background studies. COSI builds its balloon background from the transported payload, separates prompt and activation components, and compares their spectral and time behaviour with flight data [17]. DIXE likewise begins with a detailed TES payload model and asks which incident particles, production regions, and processes make the detected spectrum [18]. In our case that decomposition is especially important because source inventory and selected line-window background lead to different answers: ^{128}I dominates the reference activity, but cold-Cu isotopes dominate the selected delayed records. Designing from selected events rather than from inventory alone keeps the optimization tied to the science window.

The final component mix also shows what each instrument layer contributes. Active BGO removes most prompt and delayed events that are accompanied by shield energy. The topology test provides a smaller additional reduction while retaining nearly all focused events. The atmospheric 511 keV component survives because it is degenerate with the source in the present energy and single-photon-topology observables. Its 22.33% share at the day-15 reference state is not a failure of anticoincidence; it is a direction-dependent, full-sphere atmospheric-line background component that must be carried through the trajectory. Its separability from source emission remains to be tested with future pointing and control-region analyses.

The observing role of this telescope is complementary to INTEGRAL/SPI and COSI, which measure the diffuse bulge, disk, and broader Galactic emission [3, 4, 7, 8, 37]. The Laue-lens TES concept exchanges sky coverage for a concentrated beam and a sub-keV window. Its natural targets are compact 511 keV sources, an unresolved central component, and pointed follow-up observations. The 20 d values in Table 7 quantify that narrow-line, continuously observed reference case.

5.1 Scope and next steps

The mission values are statistical projections for the engineering detector–cryostat geometry and the synthetic, continuously on-source trajectory. They use a 50 keV offline BGO deposited-energy veto, a detector-level topology test, and all-eight-family activation in the final delayed stream. Seven families have positive delayed transports; e^- is a finite-BUILDUP zero observation, not a physical-zero claim. The veto is applied to the event-wise sum of recorded shield deposits in post-processing. It is an ideal deposited-energy threshold, not a threshold derived from a modeled BGO hardware trigger-efficiency curve. The present background budget covers only the detector–cryostat mass-model branch; background from the Laue optics and balloon platform is outside this calculation. These choices define the calculation represented by the central curve and conditional componentwise transport-counting endpoint. The latter is neither a joint 95% coverage interval nor a treatment of radiation-field, cross-section, finite- M , nuclide-mixture, calibration, or reconstruction systematics.

The atmospheric-line contribution has additional source-model limitations. The reported result uses the nominal normalization, rigidity and depth scaling, and the semi-empirical downward angular kernel of Section 3.2.2. The smooth EXPACS/PARMA continuum was not line-subtracted, so unresolved overlap with the separately transported line remains a recognized but unquantified source-model uncertainty. The mission fold also scales the total line flux while retaining the transported day-15 angular response, without a new Cosima transport in each time bin. These effects are not included in the conditional componentwise transport-counting endpoint.

The next validation can focus on the terms that matter most in the figures: increase the prompt-photon exposure that controls the conditional endpoint, repeat the all-family activation position sampling to quantify finite- M convergence, replace the reference trajectory with a flight plan and visibility history, and calibrate the energy response and veto thresholds in the final detector design. A spatial–spectral likelihood can then use the focused spot and control regions while profiling background nuisance parameters [38]. This sequence extends

the present component model without changing its source-to-geometry logic.

5.2 Conclusions

The mass-complete reference geometry shows where the focused beam, TES array, cold copper, and active shield meet. Its selected-event map identifies positrons and neutrons as the main prompt sources and cold-Cu isotopes as the delayed line-window source. Those findings lead to the final directionally graded BGO shield with 40/30/10 mm side/bottom/top coverage. At the day-15 reference state, the final-geometry full chain gives $B = 6.965 \times 10^{-3} \text{ s}^{-1}$ and $S = 9.862 \times 10^{-4} \text{ s}^{-1}$ at the top-of-atmosphere reference flux $F_0 = 10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$. Over the fixed-45°-elevation, 20 d reference trajectory, the central and conditional componentwise transport-counting results reach $Z = 15.20$ and 5.67 , corresponding to 3σ thresholds of 1.974×10^{-5} and $5.293 \times 10^{-5} \text{ ph cm}^{-2} \text{ s}^{-1}$. The result demonstrates how a particle- and material-resolved background model can be turned into a clear final geometry and a mission-scale 511 keV performance prediction.

Data and software availability

Before publication, a versioned public repository or institutional archive should provide the geometry and source definitions, random seeds, analysis and figure-generation scripts, LaTeX source, and derived validation products needed to reproduce the tables and figures in this paper.

Declarations

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